Design of Spinodal Microstructures in High-Dimensional Composition Spaces via Analytic Derivatives of CALPHAD Model Predictions

Courtney Kunselman[a,*], Doğuhan Sarıtürk[a], Siya Zhu[a], Vahid Attari[a], Raymundo Arróyave[a,b,c]

[a]*Department of Materials Science and Engineering, Texas A&M University, College Station, 77843, TX, USA*
[b]*J. Mike Walker '66 Department of Mechanical Engineering, Texas A&M University, College Station, 77843, TX, USA*
[c]*Wm Michael Barnes '64 Department of Industrial and Systems Engineering, Texas A&M University, College Station, 77843, TX, USA*

Abstract

Keywords:

1. Introduction

Spinodal decomposition is a phase separation process characterized by barrierless uphill diffusion resulting in a bicontinuous microstructure morphology with periodic oscillations in composition. Since properties arise from structure, it is important to identify regions of conditions space which may be unstable to composition fluctuations and susceptible to spontaneous phase separation when designing alloys. In some cases, spinodal decomposition is undesirable and can lead to significant hardening and embrittlement of single-phase ductile alloys [1] or provide a a pathway for the formation of brittle, deleterious phases [2]. On the other hand, spinodal and spinodal-assisted microstructures have recently been leveraged to overcome the strength-ductility tradeoff [3–5], tailor magnetic properties [6, 7], and improve thermoelectric performance [8]. In the equiatomic HfNbTiV alloy, An et al. reported lattice strains of up to 5 % in the minority BCC phase, the largest measured among refractory HEAs, demonstrating the exceptional elastic state achievable through spinodal decomposition [3].

Regardless of whether the designer is looking to exploit or avoid spinodal decomposition, such a region must first be located within a design space that is often high-dimensional. Because exhaustive experimental characterization of these large design spaces is infeasible, computational modeling approaches which can predict microstructure stability are essential for the efficient identification of spinodal regions. Recent work to this end includes high-throughput *ab initio* calculations coupled with cluster expansion to construct CALPHAD-style Gibbs energy models [9] and a novel procedure not requiring any functional form of the free energy in which the spinodal decomposition process is modeled as a nucleation event infinitesimally close to the spinodal boundary [10]. While these approaches do not require any empirical data and were well-validated with experiments, they were only demonstrated on binary systems. Showing promise for spinodal design in high-dimensional composition spaces, Koneru, Kadirvel, and Wang recently developed

*Corresponding author
Email address: cjkunselman18@tamu.edu (Courtney Kunselman)

a CALPHAD-based high-throughput microstructure stability predictive framework specifically for the multi-principal element alloy (MPEA) or high entropy alloy (HEA) design problem [11, 12]. This framework, now implemented in Pandat [13], facilitates gauge-invariant prediction and visualization of summaries of Gibbs energy surface geometry such as eigenvalues and eigenvectors of the Gibbs energy Hessian, spinodal temperatures, and initial composition fluctuations.

In this work, we start with the framework of Koneru, Kadirvel, and Wang as a foundation and demonstrate a few key extensions. First, rather than using numerical derivative approximations which can become computationally expensive as the number of components gets large, we employ the recently formalized Jansson derivative method to calculate analytic Gibbs energy Hessians in a robust and efficient manner [14]. Second, instead of relying on access to commercial databases which may not contain models for all alloy systems of interest, we use the recently developed PhaseForge package [15] to automatically construct multicomponent CALPHAD models from machine learning-based interatomic potentials (MLIP). Third, we include an estimate for the elastic energy contributions due to coherency strains which penalize phase separation and effectively shrink the miscibility gap and spinodal region. Fourth, instead of using a collection of constant core component (a type of pseudoternary) diagrams to visualize results, we employ affine projections [16] to summarize high-dimensional data on a single two-dimensional plot.

Lastly, the entire workflow is packaged in an open-source tool available through the MaterialsFramework interface [17]. Through this framework, modelers have whitebox access to the CALPHAD model, and as demonstrated in [18], they can further refine parameters to any available computational or experimental data through ESPEI [19]. The remainder of this work will dive into the theory and methodology of determining microstructure stability from the Gibbs energy description of a phase and then demonstrate our framework to investigate the stability of the BCC phase in the Hf-Nb-Ti-V system.

2. Theory

2.1. Developing Gibbs Energy Models From Universal Potentials

The stability analysis described in the following section requires a molar Gibbs energy description of the solid solution phase of interest as a function of composition and temperature. Classical CALPHAD databases are resource-intensive to develop and may not exist for all alloy systems of interest. To overcome this limitation, we employ the PhaseForge package [15], which automates the construction of CALPHAD-compatible Gibbs energy models from MLIPs as described in [20, 21]. This approach replaces computationally expensive density functional theory (DFT) calculations with MLIP evaluations that are several orders of magnitude faster while maintaining thermodynamic accuracy.

The approach is built on the use of special quasirandom structures (SQS) [22] to represent disordered solid solutions in a periodic cell. An SQS is a periodic supercell whose short-range order parameters match those of a perfectly

random alloy up to a pre-determined neighbor range, enabling the disordered state to be represented without requiring large supercells.

Following the compound energy formalism (CEF) as implemented in the Alloy Theoretic Automated Toolkit (ATAT) [23], the molar Gibbs energy of solution phase β with site fractions $\boldsymbol{y}$ at temperature T is given by

$$G_m^{\beta}(\boldsymbol{y}, T) = \left(G^{\beta}_{\mathrm{calc,nc}}(\boldsymbol{y}, T) - \sum_i x_i \, G^{\alpha_i}_{\mathrm{calc,nc}}(T) \right) + \sum_i x_i \, G^{\alpha_i}_{\mathrm{SGTE}}(T) - T S_{\mathrm{id}}(\boldsymbol{y}), \tag{1}$$

where $G^{\beta}_{\mathrm{calc,nc}}$ is the Gibbs energy of phase β computed from MLIP total energies without any configurational entropy contribution, $G^{\alpha_i}_{\mathrm{calc,nc}}$ is the corresponding quantity for pre component i in its reference crystal structure α_i, $G^{\alpha_i}_{\mathrm{SGTE}}$ is the unary Gibbs energy of component i taken from the Scientific Group Thermodata Europe (SGTE) database [24], x_i is the mole fraction component i, and $S_{\mathrm{id}}(\boldsymbol{y})$ is the ideal configurational entropy. For a single-sublattice solution phase, the ideal configurational entropy reduces to

$$S_{\mathrm{id}}(\boldsymbol{y}) = -R \sum_i x_i \ln x_i, \tag{2}$$

where R is the ideal gas constant.

For each SQS supercell, the quantity $G^{\beta}_{\mathrm{calc,nc}}$ at 0 K is taken directly from the MLIP-relaxed potential energy per atom. Structural relaxations are performed through the MaterialsFramework calculator interface [17] with a force convergence criterion of $f_{\max} < 0.001\,\mathrm{eV/Å}$. For the BCC_A2 phase in the Hf−Nb−Ti−V system, the compositional dependence of the mixing energy is captured sufficiently by static formation energies, and thermal contributions are accounted for through the SGTE unary data.

Once formation energies have been computed for SQS supercells spanning the composition space of each phase, the composition-dependent mixing contribution embedded in $G^{\beta}_{\mathrm{calc,nc}}$ is parameterized using a Redlich-Kister polynomial expansion. For a binary A−B subsystem, the excess Gibbs energy is expressed as

$$G^{\beta}_{m,\mathrm{xs}} = x_A x_B \sum_{\nu=0}^{k} L^{(\nu)}_{A,B} \, (x_A - x_B)^{\nu}, \tag{3}$$

where $L^{(\nu)}_{A,B}$ are the ν-th order binary interaction parameters determined by least-squares fitting to the MLIP-computed formation energies. Temperature-independent interaction parameters are assumed throughout. Higher-order ternary and quaternary interactions are incorporated analogously and are extracted by fitting to SQS energies at compositions spanning the multicomponent space.

The fitted interaction parameters are assembled into a thermodynamic database (TDB) file compatible with PyCalphad [25]. The resulting TDB for the BCC_A2 phase in the Hf−Nb−Ti−V system serves as the thermodynamic input for the stability analysis described below. Although pure Hf and Ti adopt the HCP structure at room temperature, the $\mathrm{BCC_{A2}}$ phase is stabilized at the concentrated compositions studied here and is the experimentally observed structure [3]; the analysis throughout this work is therefore restricted to the $\mathrm{BCC_{A2}}$ phase.

2.2. Determining Stability From the Gibbs Energy Hessian

Gibbs first introduced the concept of a thermodynamic stability limit, or spinodal, by considering composition fluctuations leading to continuous changes in fluid phases [26]. He determined that the limit of stability for such a phase relative to these fluctuations occurs where the derivatives of the chemical potential of each component with respect to the density of its corresponding component are equal to zero. That is, a system stable to all composition fluctuations experiences positive changes in all of its chemical potentials when the content of the corresponding component is increased. As Cahn explains in [27], for binary system $A - B$ where B is the dependent component, this limit of stability simplifies to the well-known expression

$$\frac{d^2 G_m^\alpha}{dx_A^2} = 0, \tag{4}$$

where G_m^α is the molar Gibbs energy for phase α and x_A is the composition of component A.

Following Gibbs and Cahn, the pioneering work of de Fontaine rigorously derived the conditions for stability for solid phases in multicomponent systems by including coherency stress contributions to the free energy in both continuum [28] and microscopic [29] contexts. In recent works, Kadirvel *et al.* provide an elegant condensed proof of these conditions using the continuum formulation [12], and Morral and Chen apply de Fontaine's findings to comprehensibly extend Gibbs' work to the solid phase, multicomponent system context. We summarize their combined insight below.

Consider solid solution phase α, an n-component phase with composition $\boldsymbol{x}$ for which we define the n-th component as the dependent composition variable. Let $\boldsymbol{\Delta}$ be a local arbitrary composition fluctuation in phase α defined for all independent composition variables. To maintain mass balance, a local composition fluctuation of $-\boldsymbol{\Delta}$ must also occur for an equivalent amount of the system. The molar Gibbs energy change of the portion of phase α affected by this composition fluctuation is given by

$$\Delta G_m^\alpha = \frac{1}{2} G_m^\alpha(\boldsymbol{x} + \boldsymbol{\Delta}) + \frac{1}{2} G_m^\alpha(\boldsymbol{x} - \boldsymbol{\Delta}) - G_m^\alpha(\boldsymbol{x}). \tag{5}$$

Using a second order Taylor series expanded around $\boldsymbol{x}$, we can approximate $G_m^\alpha(\boldsymbol{x} + \boldsymbol{\Delta})$ as

$$G_m^\alpha(\boldsymbol{x} + \boldsymbol{\Delta}) \approx G_m^\alpha(\boldsymbol{x}) + \sum_{i=1}^{n-1} \frac{\partial G_m^\alpha}{\partial x_i} \Delta_i + \frac{1}{2} \sum_{i=1}^{n-1} \sum_{j=1}^{n-1} \frac{\partial^2 G_m^\alpha}{\partial x_i \partial x_j} \Delta_i \Delta_j \tag{6}$$

and $G_m^\alpha(\boldsymbol{x} - \boldsymbol{\Delta})$ as

$$G_m^\alpha(\boldsymbol{x} - \boldsymbol{\Delta}) \approx G_m^\alpha(\boldsymbol{x}) - \sum_{i=1}^{n-1} \frac{\partial G_m^\alpha}{\partial x_i} \Delta_i + \frac{1}{2} \sum_{i=1}^{n-1} \sum_{j=1}^{n-1} \frac{\partial^2 G_m^\alpha}{\partial x_i \partial x_j} \Delta_i \Delta_j \tag{7}$$

where all derivatives are evaluated at $\boldsymbol{x}$.

Substituting Equations (6) and (7) into Equation (5) gives

$$\Delta G_m^\alpha = \sum_{i=1}^{n-1}\sum_{j=1}^{n-1}\frac{\partial^2 G_m^\alpha}{\partial x_i \partial x_j}\Delta_i\Delta_j = \boldsymbol{\Delta}^T H \boldsymbol{\Delta} \tag{8}$$

where H is the Hessian of G_m^α with respect to the independent composition variables. Assuming that all entries of the Hessian are continuous throughout the composition domain, we can claim that H is a symmetric matrix with real eigenvalues $\lambda_1, \ldots, \lambda_{n-1}$ and a corresponding set of eigenvectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{n-1}$ which form an orthonormal basis in the space defined by the independent composition variables. Thus, $\boldsymbol{\Delta}$ can be written as a linear combination of this basis:

$$\boldsymbol{\Delta} = \sum_{i=1}^{n-1}(\boldsymbol{\Delta}^T \boldsymbol{v}_i)\boldsymbol{v}_i. \tag{9}$$

Substituting Equation (9) into Equation (8) and leveraging the relationship between eigenvalues and eigenvectors ($H\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i$) gives

$$\begin{aligned}\Delta G_m^\alpha &= \left(\sum_{i=1}^{n-1}(\boldsymbol{\Delta}^T \boldsymbol{v}_i)\boldsymbol{v}_i\right)^T H \left(\sum_{i=1}^{n-1}(\boldsymbol{\Delta}^T \boldsymbol{v}_i)\boldsymbol{v}_i\right) \\ &= \left(\sum_{i=1}^{n-1}(\boldsymbol{\Delta}^T \boldsymbol{v}_i)\boldsymbol{v}_i\right)^T \left(\sum_{i=1}^{n-1}(\boldsymbol{\Delta}^T \boldsymbol{v}_i)\lambda_i\boldsymbol{v}_i\right) = \sum_{i=1}^{n-1}\lambda_i(\boldsymbol{\Delta}^T \boldsymbol{v}_i)^2.\end{aligned} \tag{10}$$

Thus, as Equation (10) illustrates, such a composition fluctuation can only lower the Gibbs energy if at least one of the eigenvalues is negative, and a zero-valued eigenvalue represents the limit of stability, or the spinodal. If only one eigenvalue, say λ_1, is negative, the system is conditionally unstable to composition fluctuations in the direction of $\boldsymbol{v}_1$, or equivalently, with large values for the projection $\boldsymbol{\Delta}^T \boldsymbol{v}_1$. More negative eigenvalues lead to more directions for $\boldsymbol{\Delta}$ to be unstable, and $n-1$ negative eigenvalues imply that the solid solution is unstable to any $\boldsymbol{\Delta}$.

As Morral and Chen [30] and Kadirvel *et al.* [12] both report, these eigenvalues and eigenvectors are not gauge invariant. That is, $\lambda_1, \ldots, \lambda_{n-1}$ and $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{n-1}$ depend on the choice of the dependent component. This is due to a combination of the coordinate axes being geometrically non-orthogonal and the change in orientation of these axes with the variation of the dependent component.

To address this issue, Kadirvel *et al.* [12] followed the approach of Singh *et al.* [31] which involves a change of basis operation transforming H from Gibbs simplex space to Cartesian space, performing the eigenanalysis in Cartesian space, and lastly transforming the resulting eigenvectors back to Gibbs space (the eigenvalues calculated in Cartesian space remain invariant under this final transformation). Singh *et al.* [31] provide an iterative procedure for constructing such an N-dimensional linear transformation, and Kadirvel *et al.* [12] explicitly provide this matrix for systems with ten components or fewer. In this work, we adopt this approach in order to calculate and visually display the distribution of gauge-invariant eigenvalues.

Lastly, for arbitrarily-complex phases modeled with internal degrees of freedom (or equivalently, for any phase with a Gibbs energy function that is multivalued in composition space), calculating the Gibbs energy Hessian requires differentiating the outputs of single-phase energy minimization calculations. These equilibrium calculations prescribe

the composition of the phase (not the system) and determine the local minimum solution (which may be metastable) under this composition constraint. For phase α in ternary system A-B-C, it is well known that the first derivative of G_m^α with respect to x_A can be written as a simple linear combination of chemical potentials

$$\frac{\partial G_m^\alpha}{\partial x_A} = \mu_A - \mu_C \tag{11}$$

where C is the dependent component. Thus, second order derivatives can be expressed as

$$\frac{\partial^2 G_m^\alpha}{\partial x_B \partial x_A} = \frac{\partial \mu_A}{\partial x_B} - \frac{\partial \mu_C}{\partial x_B}, \tag{12}$$

but calculation of Equation (12) is less trivial. In this work, we employ the recently formalized Jansson derivative method [14], an adjoint-like technique for calculating analytic derivatives of thermodynamic properties at equilibrium with respect to CALPHAD model equilibrium calculations, to efficiently calculate the derivatives of the chemical potentials in Equation (12) rather than relying on numerical approximations. All Gibbs energy Hessians are computed using PyCalphad [25] (development version commit a8de09f).

Using the procedure described above, we can employ Jansson derivatives to calculate the chemical Hessian and corresponding chemical spinodal from a CALPHAD Gibbs energy description of a solution phase. However, in many solid solutions, composition fluctuations can lead to significant elastic energy contributions due to coherency strains. These coherency strains penalize composition fluctuations and when their elastic energy contributions are added to the chemical Gibbs energy, the spinodal region shrinks. Siya: We have to clarify that we assume the coherence is "biaxial" and the system is "infinite isotropic".As Cahn shows in [27], for an infinite binary isotropic solid with an arbitrary composition fluctuation Siya: I think the equation given by Cahn works only for infinitesimal coherent composition fluctuations instead of arbitrary or finite-amplitude composition fluctuations. He dropped the higher order terms in equation(2) and wrote "for small (c-c0)" above equation(7) in his paper. , when coherency strains are considered the limit of stability given in Equation (4) becomes

$$\frac{d^2 G_m^\alpha}{dx_A^2} + \frac{2\eta_A^2 E V_m}{1-\nu} = 0 \tag{13}$$

where E is the modulus of elasticity, V_m is the molar volume, ν is the Poisson ratio, and η_A is the linear strain per unit difference in component A:

$$\eta_A = \frac{1}{a}\frac{\partial a}{\partial x_A} \tag{14}$$

where a is the lattice parameter. Extending to the multicomponent context [32], entries of the coherent Hessian H^{coh} are given by

$$H_{ab}^{coh} = \frac{\partial^2 G_m^\alpha}{\partial x_B \partial x_A} + \frac{2\eta_A \eta_B E V_m}{1-\nu} \tag{15}$$

where $\frac{\partial^2 G_m^a}{\partial x_B \partial x_A}$ is the chemical contribution as calculated in Equation (12).

In this work, we employ the bond-based lattice parameter prediction model Siya: why don't we use the lattice parameters from the relaxed SQS directly? Courtney: because we wanted an analytic model for the lattice parameter that we could easily differentiate to calculate eta. developed by Tandoc *et al.* [33] to calculate $\boldsymbol{\eta}$ and V_m. Under this model, the average lattice parameter $\bar{a}$ for a solid solution BCC phase is given by

$$\bar{a} = \frac{2\bar{d}_{111}}{\sqrt{3}} \tag{16}$$

where $\bar{d}_{111}$ is the average bond length along the $< 111 >$ close-packed direction. The average bond length is calculated through

$$\bar{d}_{111} = \sum_i \sum_j x_i x_j d_{111}^{ij} \tag{17}$$

where d_{111}^{ij} is the first nearest neighbor (FNN) bond length for an $i-j$ pair along the $< 111 >$ close-packed direction, following the framework of Tandoc *et al.* [33]. Same-species bond lengths d_{111}^{ii} are extracted from MLIP-relaxed pure BCC cells via $d_{111}^{ii} = a_i^{\mathrm{BCC}} \sqrt{3}/2$, while cross-species lengths d_{111}^{ij} $(i \neq j)$ are obtained from MLIP-relaxed B2 (CsCl-type) binary ordered cells via $d_{111}^{ij} = a_{ij}^{\mathrm{B2}} \sqrt{3}/2$. All relaxations for bond length extraction are performed using the `BondLatticeParameter` tool within MaterialsFramework [17] with the GRACE-2L-SMAX-OMAT-large universal MLIP. Using this model for BCC structures, the molar volume is then given by

$$V_m = \frac{\bar{a}^3 N_A}{2} \tag{18}$$

where N_A is Avogadro's number. To calculate derivatives of the lattice parameter for η, a dependent component must be assigned which must remain consistent throughout the calculation of the coherent Hessian. Let k be the index of the dependent component. Then

$$x_k = 1 - \sum_{i \neq k} x_i, \tag{19}$$

and

$$\bar{d}_{111} = \sum_{i \neq k} \sum_{j \neq k} x_i x_j d_{111}^{ij} + \sum_{i \neq k} x_i (1 - \sum_{j \neq k} x_j) d_{111}^{ik} + \sum_{i \neq k} (1 - \sum_{j \neq k} x_j) x_i d_{111}^{ki} + (1 - \sum_{j \neq k} x_j)^2 d_{111}^{kk}. \tag{20}$$

Differentiating Equation (20) with respect to independent composition variable x_1 gives

$$\begin{aligned}\frac{\partial \bar{d}_{111}}{\partial x_1} &= \sum_{i\neq k}(x_i d^{i1}_{111} + x_i d^{1i}_{111}) - \sum_{i\neq 1,k}(x_i d^{ik}_{111} + x_i d^{ki}_{111}) + (1 - 2x_1 - \sum_{i\neq 1,k} x_i) d^{1k}_{111} \\ &+ (1 - 2x_1 - \sum_{i\neq 1,k} x_i) d^{k1}_{111} - 2(1 - \sum_{i\neq k} x_i) d^{kk}_{111} \\ &= 2\left(\sum_{i\neq k} x_i d^{i1}_{111} - \sum_{i\neq 1,k} x_i d^{ik}_{111} + (x_k - x_1) d^{1k}_{111} - x_k d^{kk}_{111}\right).\end{aligned} \tag{21}$$

Differentiating Equation (16) and plugging in the result from Equation (21), we have

$$\frac{\partial \bar{a}}{\partial x_1} = \frac{4}{\sqrt{3}}\left(\sum_{i\neq k} x_i d^{i1}_{111} - \sum_{i\neq 1,k} x_i d^{ik}_{111} + (x_k - x_1) d^{1k}_{111} - x_k d^{kk}_{111}\right). \tag{22}$$

Lastly, the elastic constants C_{11}, C_{12}, and C_{44} were computed for each composition of interest using the `CubicElasticConstantsAnalyzer` within the MaterialsFramework package [17]. A $4 \times 4 \times 4$ BCC SQS supercell was first relaxed using the GRACE-2L-SMAX-OMAT-large universal MLIP with unconstrained cell shape to a force convergence of $f_{\max} < 0.01\,\mathrm{eV/Å}$. Three finite strain modes were then applied to the relaxed cell: a uniform volumetric distortion to extract the bulk modulus B, an orthorhombic distortion to isolate $C' = (C_{11} - C_{12})/2$, and a monoclinic shear to determine C_{44}. Polycrystalline-averaged elastic properties were then obtained via the Voigt-Reuss-Hill (VRH) scheme, yielding the macroscopically isotropic Young's modulus E and Poisson's ratio ν. The biaxial modulus $M = E/(1-\nu)$ appearing in the coherent Heissian above was derived directly from these VRH averages. Siya: Are the Young's moduli and Poisson's ratio assumed isotropic? The Cahn's model requires the isotropy. How does materialsframework deal with that? For bcc structure the isotropy is described as $A = \frac{2C_{44}}{C_{11}-C_{12}}$ and is isotropic when $A = 1$. For Nb, given by "Elastic Constants of Niobium from 4.2 to 300K", K. Carroll, J. Appl. Phys., 1965., at 300K, the C_{11} is $24.56 \times 10^{11} dyn/cm^2$, the C_{12} is $13.87 \times 10^{11} dyn/cm^2$, and the C_{44} is $2.930 \times 10^{11} dyn/cm^2$, $A = 0.548$. The Young's moduli and Poisson ratio calculated based on the experimental data are:

Table 1: Direction-dependent Young's modulus and Poisson's ratio of single-crystal Nb at 300 K, calculated from the experimental elastic constants $C_{11} = 245.6$ GPa, $C_{12} = 138.7$ GPa, and $C_{44} = 29.30$ GPa.

Loading direction	Transverse direction	E (GPa)	ν
[100]	[010]	145.5	0.361
[110]	[1$\bar{1}$0]	93.2	0.591
[110]	[001]	93.2	0.231
[111]	[1$\bar{1}$0]	83.2	0.420

How we decide the E and ν in Equation 12?

2.3. *Verifying Microstructure Instability with Phase-Field Modeling*

While the eigenanalysis of the Gibbs energy Hessian described in Section 2.2 identifies *whether* a composition is spinodally unstable and in *which* compositional direction decomposition is favored, it does not reveal the temporal

evolution of the resulting microstructure. To bridge this gap, we couple the CALPHAD-derived thermodynamics directly to a phase-field simulation based on the Cahn-Hilliard equation [27, 34].

2.3.1. Free Energy Functional and Governing Equations

For an n-component alloy described by $n-1$ independent composition fields $x_i(\boldsymbol{r}, t)$, $i = 1, \ldots, n-1$, with the n-th component serving as the reference (dependent variable), the total free energy functional is

$$\mathcal{F}[\boldsymbol{x}] = \int_\Omega \left[G_m^\alpha(\boldsymbol{x}) + \sum_{i=1}^{n-1} \frac{\kappa_i}{2} \left| \nabla x_i \right|^2 \right] dV, \tag{23}$$

where $G_m^\alpha(\boldsymbol{x})$ is the molar Gibbs energy of phase α (the CALPHAD-derived quantity described in Section 2.1), and κ_i (J m^2 mol^{-1}) is the gradient energy coefficient for field i that penalizes sharp compositional interfaces. The time evolution of each composition field is governed by the Cahn-Hilliard equation:

$$\frac{\partial x_i}{\partial t} = \nabla \cdot \sum_{j=1}^{n-1} M_{ij}(\boldsymbol{x}) \, \nabla \mu_j^{\mathrm{tot}}, \qquad i = 1, \ldots, n-1, \tag{24}$$

where $\boldsymbol{M}(\boldsymbol{x})$ is the composition-dependent mobility tensor (m^5 J^{-1} s^{-1}) and μ_j^{tot} is the variational chemical potential:

$$\mu_j^{\mathrm{tot}} = \frac{\delta \mathcal{F}}{\delta x_j} = \underbrace{\frac{\partial G_m^\alpha}{\partial x_j}}_{\mu_j - \mu_n} - \kappa_j \nabla^2 x_j. \tag{25}$$

The bulk driving force $\mu_j - \mu_n = \partial G_m^\alpha / \partial x_j$ is precisely the relative chemical potential already established in Equation (11). To prevent diffusion from diverging as any component approaches a pure-element limit, we employ a degenerate mobility,

$$M_{ij}(\boldsymbol{x}) = \mathring{M}_{ij} \prod_{k=1}^{n} x_k, \tag{26}$$

where $\mathring{M}_{ij}$ is the composition-independent (bare) mobility matrix. The product over all n mole fractions vanishes whenever any $x_k \to 0$, enforcing physical boundary behaviour.

Equation (24) is solved numerically on a uniform $N_g \times N_g$ grid (domain size L_x) using a pseudo-spectral semi-implicit scheme [35]. Diagonal mobility terms are treated implicitly (providing unconditional damping of high-wavenumber modes) while off-diagonal cross-terms are treated explicitly:

$$\hat{x}_i^{n+1} = \frac{\hat{x}_i^n + \Delta t \, k^2 \sum_j M_{ij}^{\mathrm{eff}} \, \hat{\mu}_{j,\mathrm{bulk}}^n - \Delta t \, k^4 \sum_{j \neq i} M_{ij}^{\mathrm{eff}} \, \kappa_j \, \hat{x}_j^n}{1 + \Delta t \, k^4 \, M_{ii}^{\mathrm{eff}} \, \kappa_i}, \tag{27}$$

where $\hat{(\cdot)}$ denotes the discrete Fourier transform, k^2 is the squared wavenumber, and $M_{ij}^{\mathrm{eff}} = \mathring{M}_{ij} \langle \prod_k x_k \rangle$ is the spatially-averaged effective mobility. The stable time step is estimated from the Hessian of the free energy at the mean composition $\boldsymbol{x}_0$:

$$\Delta t_{\mathrm{stable}} = \frac{\alpha_{\mathrm{s}} \, \Delta x^2}{\max_i(\mathring{M}_{ii}) \, |\lambda_{\max}(H)|}, \tag{28}$$

where $\alpha_{\mathrm{s}} < 1$ is a safety factor, $\Delta x = L_x / N_g$, and $|\lambda_{\max}(H)|$ is the largest absolute eigenvalue of the Gibbs energy Hessian $H_{ij} = \partial^2 G_m^\alpha / \partial x_i \partial x_j$ evaluated in a neighbourhood of $\boldsymbol{x}_0$.

2.3.2. Eyre Convex-Splitting Stabilisation

The semi-implicit scheme of Equation (27) treats the gradient energy term $\kappa_i k^4$ implicitly, which unconditionally damps high-wavenumber oscillations arising from interfacial stiffness. However, within the spinodal region the diagonal Hessian elements $H_{ii} = \partial^2 G_m^\alpha / \partial x_i^2$ are *negative*, and the corresponding bulk contribution $M_{ii} H_{ii} k^2$ enters the numerator with a destabilising sign. For large $|\Delta t H_{ii}|$ this explicit term can cause unbounded growth and numerical blowup, regardless of the spatial resolution.

Following Eyre [**?**], we apply a convex-splitting decomposition of the molar Gibbs energy G_m^α into a *contracting* (convex) part G^+ and an *expanding* (concave) part G^-,

$$G_m^\alpha(\boldsymbol{x}) = G^+(\boldsymbol{x}) - G^-(\boldsymbol{x}), \qquad G^\pm \text{ convex}, \tag{29}$$

and treat G^+ implicitly and G^- explicitly in time. For a quadratic approximation to the bulk free energy at the mean composition $\boldsymbol{x}_0$, a sufficient convex-splitting for component i is

$$G_i^+ = \tfrac{A_i}{2} x_i^2, \qquad G_i^- = \tfrac{A_i}{2} x_i^2 - G_m^\alpha(\boldsymbol{x}), \tag{30}$$

where the *stabilisation constant* A_i must satisfy

$$A_i \;\geq\; \max(0,\, -H_{ii}(\boldsymbol{x}_0)) \tag{31}$$

to ensure that $G_i^+ - \frac{A_i}{2} x_i^2$ remains convex [**? ?**]. With this splitting, the update rule for field i in Fourier space becomes

$$\hat{x}_i^{n+1} = \frac{\hat{x}_i^n + \Delta t\, k^2 \sum_j M_{ij}^{\text{eff}}\, \hat{\mu}_{j,\text{bulk}}^n - \Delta t\, k^2\, M_{ii}^{\text{eff}}\, A_i\, \hat{x}_i^n - \Delta t\, k^4 \sum_{j \neq i} M_{ij}^{\text{eff}}\, \kappa_j\, \hat{x}_j^n}{1 + \Delta t\, k^2\, M_{ii}^{\text{eff}}\, A_i + \Delta t\, k^4\, M_{ii}^{\text{eff}}\, \kappa_i}, \tag{32}$$

where the diagonal chemical potential term $M_{ii}^{\text{eff}}\, \hat{\mu}_{i,\text{bulk}}^n$ and off-diagonal bulk cross-terms $\sum_{j \neq i} M_{ij}^{\text{eff}}\, \hat{\mu}_{j,\text{bulk}}^n$ both enter at order k^2, while the off-diagonal gradient cross-terms enter at order k^4, consistent with Equation (27). The denominator is strictly positive for any $\Delta t > 0$, $A_i \geq 0$, $\kappa_i > 0$, making the scheme *unconditionally stable*: the solution cannot grow without bound regardless of the chosen time step.

Role of the stabilisation constant.. The choice of A_i controls the balance between stability and accuracy. To see this, note that the *effective* bulk driving force seen by field i in the numerator of Equation (32) is

$$H_{ii}^{\text{eff}} = H_{ii}(\boldsymbol{x}_0) + A_i. \tag{33}$$

Three regimes arise:

1. $A_i = |H_{ii}|$ **(minimum required).** $H_{ii}^{\text{eff}} = 0$: the linear spinodal driving force is exactly neutralised in the explicit part, and decomposition is driven purely by the *nonlinear* deviation of $\mu_{i,\text{bulk}}$ from its linearisation about $\boldsymbol{x}_0$. This is the theoretically optimal choice: the scheme is unconditionally stable and introduces no artificial damping of the spinodal growth rate.

2. $A_i > |H_{ii}|$ **(over-stabilised).** $H_{ii}^{\text{eff}} > 0$: the explicit update treats component i as if it were thermodynamically *stable*, actively opposing spinodal growth at every time step. The scheme remains unconditionally stable and will not blow up for any Δt, but the linear spinodal growth rate is artificially suppressed. Decomposition still eventually occurs, driven by the nonlinear chemical potential, but at a rate slower than the true physical dynamics.
3. $A_i < |H_{ii}|$ **(under-stabilised).** $H_{ii}^{\text{eff}} < 0$: the explicit term retains a net destabilising contribution. The scheme may blow up for large Δt and is only conditionally stable, recovering the classical CFL restriction.

Practical determination of A_i.. The condition Equation (31) is a lower bound, not a prescription. In practice, selecting A_i requires care because the diagonal Hessian $H_{ii}(\boldsymbol{x})$ is *not constant*: it depends on composition, temperature, and the alloy system. A value of A_i that is sufficient at the nominal composition $\boldsymbol{x}_0$ may become inadequate as the field evolves away from $\boldsymbol{x}_0$ during decomposition. Several strategies are possible.

1. **Point evaluation at $\boldsymbol{x}_0$.** The simplest approach sets $A_i = \max(0, -H_{ii}(\boldsymbol{x}_0))$. This is exact at the initial composition but may be insufficient once spinodal decomposition drives the local composition toward the equilibrium tie-line endpoints, where $|H_{ii}|$ can be substantially larger.
2. **Neighborhood scan (recommended).** Evaluate $H_{ii}(\boldsymbol{x})$ on a coarse grid spanning a window $[\boldsymbol{x}_0 \pm \delta]$ around the nominal composition, and take $A_i = \max_{\boldsymbol{x}\in\text{scan}} \max(0, -H_{ii}(\boldsymbol{x}))$. This anticipates the composition excursions during decomposition without evaluating the full spinodal region. In the present implementation, a window $\delta = 0.30$ (mole fraction) is used by default, sampled on a 20^d grid where d is the number of independent composition variables.
3. **Full spinodal scan.** Evaluate H_{ii} everywhere the diagonal element is negative (i.e. inside the spinodal boundary) and set A_i to the global minimum of H_{ii}. This guarantees unconditional stability for all reachable compositions but may over-stabilise the early dynamics.
4. **Adaptive A_i.** Recompute $A_i(t) = \max(0, -H_{ii}(\bar{\boldsymbol{x}}(t)))$ periodically from the current spatial mean $\bar{\boldsymbol{x}}(t)$ as the simulation progresses. This is theoretically optimal: A_i tracks the evolving composition and remains as close to $|H_{ii}|$ as possible at every stage, neither over-stabilising the early spinodal nor under-stabilising the late coarsening.

The sensitivity of A_i to composition can be significant. As a concrete example, for the binary $Nb_{50}V_{50}$ system at 500 °C, the diagonal Hessian element $H_{\text{Nb}}(\boldsymbol{x})$ takes the value $-7239\,\text{J mol}^{-1}$ at the nominal composition $x_{\text{Nb}} = 0.50$, but reaches $-19\,840\,\text{J mol}^{-1}$ near $x_{\text{Nb}} = 0.75$, an increase of nearly 2.7×. Relying solely on the point evaluation at $\boldsymbol{x}_0$ therefore underestimates the required stabilisation constant by that factor, causing the scheme to lose unconditional stability as the field composition evolves.

This sensitivity is also alloy- and temperature-dependent. In the ternary $Nb_{50}Hf_5V_{45}$ system, H_{Nb} varies from $0\,\text{J mol}^{-1}$ (neutrally stable) near 750 °C to $-5547\,\text{J mol}^{-1}$ at 600 °C and more negative values at lower temperatures, while H_{Hf} remains positive at all temperatures studied here. Consequently, the required A_i differs component by component, alloy by alloy, and temperature by temperature; there is no universal constant that applies across systems.

Per-component stabilisation.. A key feature of the present implementation is that A_i is set *independently for each component*. Elements with $H_{ii} > 0$ everywhere in the scan window (e.g. Hf in Nb-Hf-V at 600 °C, with $H_{\mathrm{Hf}} \approx 87\,720\,\mathrm{J\,mol^{-1}}$) receive $A_i = 0$ and contribute no extra stabilisation. Spinodally unstable elements (e.g. Nb, with $H_{\mathrm{Nb}} = -5547\,\mathrm{J\,mol^{-1}}$ at 600 °C) receive the minimum A_i needed to satisfy Equation (31) over the composition scan. This per-component assignment avoids introducing unnecessary artificial damping on stable species, which would slow their diffusional response and distort the coupled composition dynamics.

Interpolant boundary effects.. A practical issue distinct from numerical stability arises when composition field values overshoot the boundary of the pre-computed thermodynamic table. The surrogate interpolant is constructed on a finite composition grid; if x_i^n at any spatial point lies outside that grid, the interpolated chemical potential is undefined (typically returning `NaN`). Because the update Equation (32) is evaluated in Fourier space, a single `NaN` propagates instantaneously to every grid point, halting the simulation. This is *not* a stability failure of the Eyre scheme; it is a thermodynamic evaluation error, and it cannot be resolved by adjusting Δt or A_i.

The correct remedy is to evaluate the chemical potential on a *clamped copy* of the composition field, $\boldsymbol{x}^{\mathrm{eval}} = \mathrm{clamp}(\boldsymbol{x}^n, \epsilon, 1-\epsilon)$, rescaled to satisfy the simplex constraint, while updating the PDE field $\boldsymbol{x}^{n+1}$ from the unclamped $\boldsymbol{x}^n$. The stored composition field is never modified, so mass conservation is preserved exactly. The clamped evaluation is only invoked for points that would otherwise lie outside the interpolant domain; in the interior of the spinodal region these corrections are negligible.

Consequence for the time-step safety factor.. Because the Eyre scheme is unconditionally stable, it does not blow up even when the time step Δt exceeds the classical stability limit $\Delta t_{\mathrm{stable}}$ of Equation (28). The safety factor α_{s} in Equation (28) therefore controls *accuracy* rather than *stability*: larger α_{s} gives a coarser temporal resolution of the early-stage spinodal amplification and coarsening dynamics, but will not produce a divergent solution. In practice, values $\alpha_{\mathrm{s}} \leq 1$ give accurate reproduction of the linear spinodal growth rate, while values $\alpha_{\mathrm{s}} \gg 1$ can be used to fast-forward to late-stage microstructural states at the cost of temporal fidelity in the early regime.

2.3.3. CALPHAD-Coupled Surrogate for Thermodynamic Driving Forces

Direct evaluation of pycalphad at every spatial grid point and every time step is computationally prohibitive (each equilibrium call requires an iterative minimization). Instead, all thermodynamic quantities are pre-computed on a dense composition grid spanning the relevant simplex, stored in an HDF5 file, and replaced by smooth surrogate functions that are evaluated in $O(\log N)$ time during simulation.

At each grid point $\boldsymbol{x}^{(k)}$, three quantities are evaluated once using pycalphad's `IsolatedPhase` framework, which constrains every calculation to the single-phase $\mathrm{BCC_{A2}}$ free energy surface: the molar Gibbs energy $G_m^{\alpha,(k)}$ ($\mathrm{J\,mol^{-1}}$); the absolute chemical potential $\mu_i^{(k)}$ of each element i ($\mathrm{J\,mol^{-1}}$); and the Jansson second derivative $\Lambda_{ij}^{(k)} = \partial\mu_i/\partial x_j$ [14] ($\mathrm{J\,mol^{-1}}$), which is the analytic derivative of the chemical potential with respect to composition at single-phase

equilibrium. All three datasets are stored in a structured HDF5 file and loaded once at the start of a simulation run; no further thermodynamic calculations are performed during time integration.

The surrogate is constructed using composition-appropriate interpolation backends: a cubic spline for binary systems ($n = 2$), a C^1-smooth Clough-Tocher triangular interpolant [36] for ternary systems ($n = 3$), and a two-stage thin-plate-spline RBF-to-regular-grid scheme [37] for quaternary and higher-order systems ($n \geq 4$). All three methods share the same surrogate class and are used identically for G_m^α, μ_i, and Λ_{ij}.

2.3.4. Derivative Methods for the Bulk Chemical Potential

Computing the bulk driving force $\mu_j - \mu_n = \partial G_m^\alpha / \partial x_j$ and the Hessian H_{ij} from the surrogate can be done in three ways, selectable by the user. The choice matters because the Hessian controls both the spinodal wavelength $\lambda^* = 4\pi\sqrt{\kappa_i/|H_{ij}|}$ and the stable time step (Eq. 28), so derivative errors propagate directly into microstructural length and time scales.

Method I: Finite differences on G_m^α (`fd`).. The chemical potential and Hessian are obtained numerically from the G_m^α surrogate by central finite differences:

$$\left(\mu_j - \mu_n\right)^{\mathrm{fd}} = \frac{G_m^\alpha(\boldsymbol{x} + \varepsilon \boldsymbol{e}_j) - G_m^\alpha(\boldsymbol{x} - \varepsilon \boldsymbol{e}_j)}{2\varepsilon} + O(\varepsilon^2), \tag{34}$$

and similarly for H_{ij} using the standard second-order stencil, where $\boldsymbol{e}_j$ is the unit vector in the j-th composition direction and $\varepsilon = 10^{-4}$. This approach requires only the G_m^α table; no chemical potential data need be stored. However, truncation error compounds with the smoothing bias of the surrogate, and errors in H_{ij} are amplified near the simplex boundaries where the ideal-entropy contribution $-RT\sum_i x_i \ln x_i$ makes G_m^α highly curved.

Method II: Direct Jansson interpolation (`jansson`).. Under the simplex constraint $x_n = 1 - \sum_{i=1}^{n-1} x_i$, the Gibbs-Duhem equation gives [38]:

$$\mu_j - \mu_n = \left.\frac{\partial G_m^\alpha}{\partial x_j}\right|_{\text{simplex}}, \tag{35}$$

and the Hessian element is (from Equation (12)):

$$H_{ij} = \left.\frac{\partial^2 G_m^\alpha}{\partial x_i \, \partial x_j}\right|_{\text{simplex}} = \Lambda_{ij} - \Lambda_{nj}, \tag{36}$$

where $\Lambda_{ij} = \partial \mu_i / \partial x_j$ is the Jansson second derivative [14] pre-computed and stored in the HDF5 table. Separate surrogate functions are constructed for each $\mu_j - \mu_n$ (from Eq. 35) and each H_{ij} (from Eq. 36) using identical backends. This method is free of finite-difference truncation error and satisfies the Gibbs-Duhem relation exactly at every queried composition. For an n-component system, it requires $(n-1)$ additional surrogates for the chemical potentials and $(n-1)^2$ for the Hessian components, a total of $(n-1)n$ interpolants beyond the G_m^α surface, all constructed once at simulation startup.

Method III: Hybrid: Jansson μ, finite-difference H (`hybrid`).. The bulk driving force $\mu_j - \mu_n$ is obtained from the Jansson surrogates (Eq. 35), while the Hessian H_{ij} is obtained by finite differences on G_m^α (Eq. 34). This combination is appropriate when the Jansson derivative dataset (`dMU`) is unavailable or when pycalphad's numerical Hessian data are unreliable near a phase boundary, while still providing thermodynamically consistent chemical potentials for the time-evolution equation.

2.3.5. *Validation of Derivative Methods*

Figure 1 compares the three modes against raw CALPHAD data for the Nb-V BCC_{A2} binary system at $T = 500\,°C$ as a function of the vanadium mole fraction x_V. Panel (a) confirms that the molar Gibbs energy G_m^α is identical across all modes, since it is always evaluated from the same G_m^α surrogate. Panel (b) shows the relative chemical potential $\mu_{Nb} - \mu_V$: the `fd` mode deviates visibly from the CALPHAD reference (RMSE = $302\,J\,mol^{-1}$) while `jansson` and `hybrid` reproduce the stored values exactly (RMSE = $0\,J\,mol^{-1}$). Panel (c) shows the most striking difference: the Hessian $\partial^2 G_m^\alpha / \partial x_{Nb}^2$ is displayed with the y-axis clipped to $\pm 150\,kJ\,mol^{-1}$ (the ideal-entropy divergence at $x_V \to 0$ or 1 is annotated); the `fd` mode yields an RMSE of $442\,kJ\,mol^{-1}$ relative to the CALPHAD reference, entirely due to truncation error amplification near the simplex boundaries, whereas `jansson` matches exactly. Since the spinodal decomposition wavelength scales as $\lambda^* \propto |H_{ij}|^{-1/2}$, Hessian errors of this magnitude would translate into erroneous microstructural length-scale predictions; the `jansson` method eliminates this source of error.

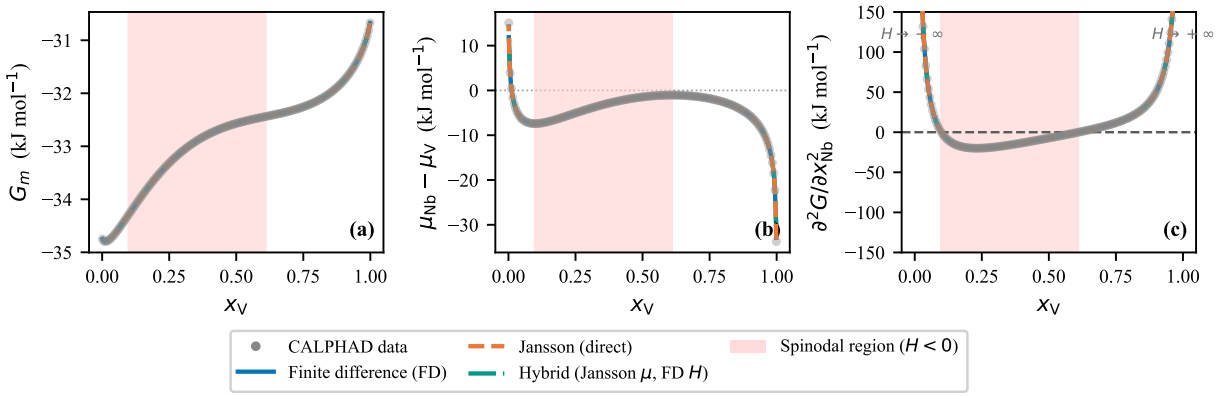


Figure 1: Comparison of the three derivative methods for the Nb-V BCC_{A2} system at $T = 500\,°C$. Grey markers: raw CALPHAD data (60% transparent to reveal underlying curves). (a) Molar Gibbs energy G_m^α: all three modes are identical (shared surrogate). (b) Relative chemical potential $\mu_{Nb} - \mu_V$: `fd` deviates by RMSE = $302\,J\,mol^{-1}$; `jansson` and `hybrid` are exact. (c) Hessian $\partial^2 G_m^\alpha / \partial x_{Nb}^2$ (clipped to $\pm 150\,kJ\,mol^{-1}$; divergence at the pure-element limits is annotated): `fd` RMSE = $442\,kJ\,mol^{-1}$; `jansson` is exact. The shaded pink band marks the spinodal region ($H < 0$) identified from the `jansson` Hessian.

To confirm that point-wise derivative agreement translates into equivalent field evolution, we ran two identical Cahn-Hilliard simulations, one each for the `fd` and `jansson` modes, on the Nb-V BCC_{A2} binary at $T = 500\,°C$ using a 256×256 periodic domain of side $L = 1\,\mu m$, $\kappa = 1 \times 10^{-12}\,J\,m^{-1}$, $M = 5 \times 10^{-21}\,m^3\,s\,kg^{-1}$, and 80 000 time steps of $\Delta t = 0.01\,s$ ($t = 800\,s \approx 26\,\tau$, where $\tau \approx 31\,s$ is the spinodal growth time constant at the dominant wavenumber

Table 2: Summary of derivative methods for bulk chemical potentials and Hessian in the CALPHAD-coupled Cahn-Hilliard model. RMSE values are for the Nb-V BCC_{A2} binary at $T = 500\,°C$ ($n = 200$ composition grid points). Bold: exact agreement with CALPHAD data.

Mode	$\mu_j - \mu_n$ source	H_{ij} source	HDF5 data required	$RMSE_\mu$ (J mol^{-1})	$RMSE_H$ (J mol^{-1})
`fd`	FD on G_m^α (Eq. 34)	FD on G_m^α	`/GM`	302	442 219
`jansson`	Eq. 35	Eq. 36	`/GM`, `/MU`, `/dMU`	**0**	**0**
`hybrid`	Eq. 35	FD on G_m^α	`/GM`, `/MU`	**0**	442 219

$k^* = \sqrt{|H_{\min}|/4\kappa}$). Both runs were seeded identically; the only variable was the derivative method used to evaluate the chemical-potential driving force $\delta F/\delta c$. Figure 2 presents the final x_V composition fields and the absolute per-pixel difference map. After 800 s of evolution the field is fully phase-separated, with well-coarsened Nb-rich ($x_V \approx 0.02$) and V-rich ($x_V \approx 0.82$) domains of characteristic size ~ 200 nm. The two micrographs are visually identical; `fd` shows a small residual relative to `jansson` (RMSE $= 1.3 \times 10^{-5}$, max $|\Delta x_V| = 2.4 \times 10^{-4}$) concentrated at domain interfaces where the composition gradient is steepest and FD truncation error is most amplified. This tiny residual does not affect the phase morphology or domain statistics, confirming that both derivative modes are effectively equivalent for the binary system. Differences may grow for higher-order systems where scattered-data interpolants are used; those cases are addressed in **??**.

2.3.6. Elastic Coupling via Coherency Strain

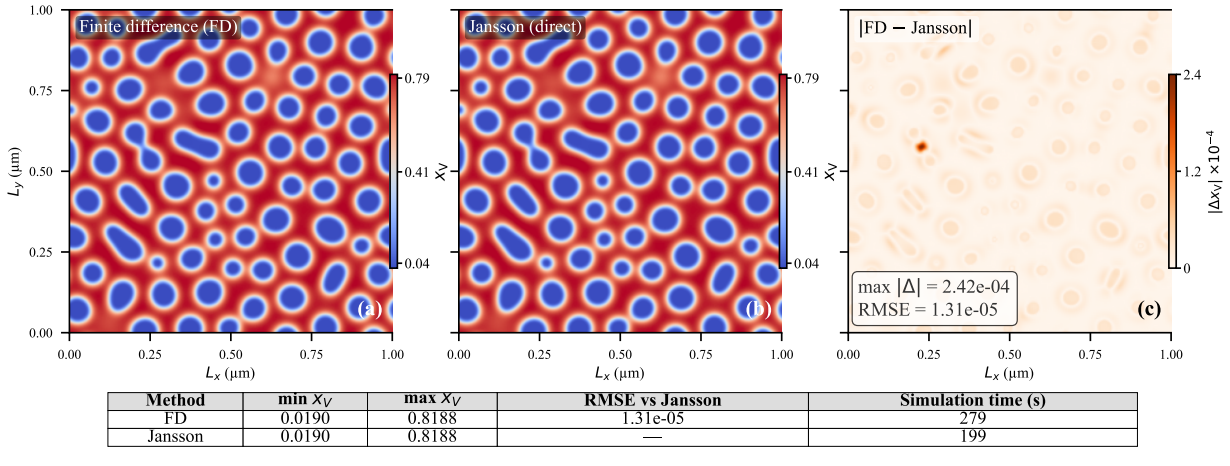
Coherency strains arising from composition-dependent lattice mismatch contribute an elastic free energy that supplements the chemical driving force. Following Gururajan and Abinandanan [**?**], the total chemical potential entering Equation (24) is augmented by an elastic term:

$$\mu_j^{\text{tot}} = \mu_j^{\text{ch}} + \mu_j^{\text{el}}, \tag{37}$$

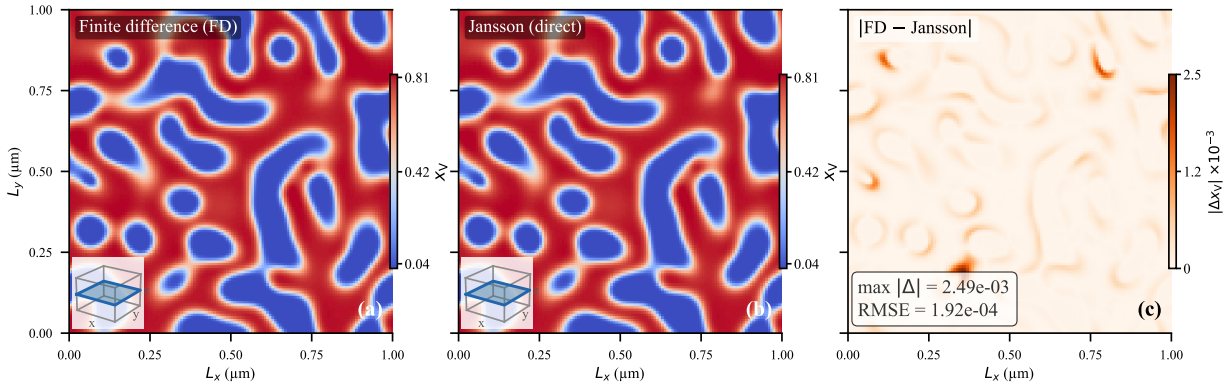
where $\mu_j^{\text{ch}} = \partial G_m^\alpha/\partial x_j - \kappa_j \nabla^2 x_j$ is the chemical contribution (Equation (25)) and μ_j^{el} is the elastic contribution derived from the variation of the elastic strain energy with respect to composition.

Mechanical equilibrium is solved at each elastic update step using the Lippmann-Schwinger spectral scheme of Moulinec and Suquet [**?**], which reformulates the Navier equation as a Fredholm integral equation and solves it iteratively in Fourier space. A homogeneous reference stiffness $\mathbf{C}^0$ is chosen to accelerate convergence, and the spatially varying stiffness field $\mathbf{C}(\boldsymbol{r})$ is treated as a perturbation.

Eigenstrain and elastic chemical potential.. The elastic state of the microstructure is determined by the eigenstrain field $\boldsymbol{\varepsilon}^*(\boldsymbol{r})$, which encodes the stress-free lattice mismatch between the local composition and a reference state. Two formulations are implemented in the present framework, summarized in Table 3.



Method	min x_V	max x_V	RMSE vs Jansson	Simulation time (s)
FD	0.0190	0.8188	1.31e-05	279
Jansson	0.0190	0.8188	—	199



Method	min x_V	max x_V	RMSE vs Jansson	Simulation time (s)
FD	0.0187	0.8320	1.92e-04	19225
Jansson	0.0187	0.8320	—	22733

Figure 2: Derivative-method comparison (`fd` vs. `jansson`) for Nb-V BCC_{A2} at $T = 500\,°C$ ($c_0 = x_V = 0.50$, $L = 1\,\mu m$). *Top row* (2D, 256×256 grid, $t = 800\,s$, 80 000 steps): (a) finite-difference and (b) Jansson direct composition fields x_V; (c) absolute difference map $|x_V^{fd} - x_V^{jansson}|$ (RMSE $= 1.31 \times 10^{-5}$, max $= 2.42 \times 10^{-4}$); table: composition statistics and wall times (FD: 279 s; Jansson: 199 s). *Bottom row* (3D, 64^3 grid, $t = 800\,s$, 5 000 steps, $z = L/2$ mid-plane slice): (a) finite-difference, (b) Jansson, (c) absolute difference; discrepancies remain confined to domain interfaces in both 2D and 3D, confirming method equivalence independent of dimensionality.

In the classical model (`fixed_vegard`), the eigenstrain is linear in the composition fields with fixed Vegard coefficients $\alpha_k = \partial \ln a / \partial x_k$ evaluated at the midpoint between the tie-line endpoints. The elastic chemical potential follows directly from the variation of the elastic energy with respect to x_k: $\mu_k^{el} = -\boldsymbol{\sigma} : \boldsymbol{\Lambda}_k / N_V$, where $\boldsymbol{\Lambda}_k = \alpha_k \mathbf{I}$ is the Vegard tensor for element k. The reference stiffness is taken as the arithmetic mean of the endpoint stiffness tensors, $\mathbf{C}^0 = \frac{1}{2}(\mathbf{C}_{mat} + \mathbf{C}_{ppt})$, and the spatially varying stiffness is interpolated between the two endpoints using a smooth switching function $h(x_{ref})$.

The composition-dependent model (`local_vegard`) removes the assumption of fixed α_k by computing the eigenstrain directly from the MLIP-derived SQS lattice parameter $a(\boldsymbol{x})$ interpolated across the full composition space. The eigenstrain at each point is the volumetric mismatch relative to the parent composition: $\boldsymbol{\varepsilon}^*(\boldsymbol{r}) = [a(\boldsymbol{x}(\boldsymbol{r})) - a_0]/a_0$, which vanishes identically in the initial uniform microstructure and grows only as phase separation develops. This is

Table 3: Comparison of the two eigenstrain models available in the CALPHAD-coupled phase-field solver. α_k denotes the Vegard coefficient for element k, $a(\boldsymbol{x})$ the SQS lattice parameter at composition $\boldsymbol{x}$, $a_0 = a(\boldsymbol{x}_0)$ the reference lattice parameter at the parent composition, $\boldsymbol{\sigma}$ the Cauchy stress, and N_V the molar density. All models use the Moulinec-Suquet spectral scheme with homogeneous reference stiffness $\mathbf{C}^0$; the stiffness contrast correction is neglected in both cases, which is standard practice when the contrast is moderate.

	Classical (`fixed_vegard`)	**Composition-dependent** (`local_vegard`)
Eigenstrain field	$\boldsymbol{\varepsilon}^*(\boldsymbol{r}) = \sum_k \alpha_k \, x_k(\boldsymbol{r}) \, \mathbf{I}$	$\boldsymbol{\varepsilon}^*(\boldsymbol{r}) = \dfrac{a(\boldsymbol{x}(\boldsymbol{r})) - a_0}{a_0} \, \mathbf{I}$
Elastic μ_k^{el}	$-\boldsymbol{\sigma}(\boldsymbol{r}) : \boldsymbol{\Lambda}_k \, / \, N_V$	$-\alpha_k(\boldsymbol{x}_0) \, \mathrm{tr}(\boldsymbol{\sigma}(\boldsymbol{r})) \, / \, N_V$
Reference stiffness $\mathbf{C}^0$	$\frac{1}{2}(\mathbf{C}_{\mathrm{mat}} + \mathbf{C}_{\mathrm{ppt}})$	$\mathbf{C}(\boldsymbol{x}_0)$ from SQS interpolant
Stiffness field $\mathbf{C}(\boldsymbol{r})$	$(1-h) \, \mathbf{C}_{\mathrm{mat}} + h \, \mathbf{C}_{\mathrm{ppt}}$	SQS interpolant evaluated pointwise at $\boldsymbol{x}(\boldsymbol{r})$
α_k evaluation	Fixed at tie-line midpoint (step 1b)	$\partial \ln a / \partial x_k$ at $\boldsymbol{x}_0$ via finite difference on SQS interpolant
Initial $\boldsymbol{\varepsilon}^*$ (uniform $\boldsymbol{x}_0$)	$\sum_k \alpha_k \, x_{k,0} \, \mathbf{I} \neq \mathbf{0}$	$\mathbf{0}$ (vanishes exactly)

physically correct: an undecomposed single-phase alloy carries no coherency stress. The classical model, by contrast, assigns a nonzero eigenstrain proportional to $\boldsymbol{x}_0$ everywhere in the initial field. The reference stiffness $\mathbf{C}^0 = \mathbf{C}(\boldsymbol{x}_0)$ is evaluated from the SQS interpolant at the parent composition, capturing the correct elastic environment of the parent phase, while the spatially varying stiffness $\mathbf{C}(\boldsymbol{r})$ is evaluated pointwise from the same interpolant. The elastic chemical potential uses $\alpha_k(\boldsymbol{x}_0)$, the Vegard coefficient at the parent composition computed by finite difference on the SQS interpolant, rather than a midpoint value, which is a first-order approximation that avoids per-point gradient evaluations on the scattered interpolant.

In both models, the stiffness-contrast correction ($\frac{1}{2}\delta\mathbf{C} : \boldsymbol{\varepsilon}^{\mathrm{el}} : \boldsymbol{\varepsilon}^{\mathrm{el}}$) is neglected, as is standard in spectral phase-field codes when the elastic contrast between the two phases is moderate [**?**]. The mechanical equilibrium is re-solved every n_{el} time steps and the resulting elastic chemical potential is cached between solves to reduce computational cost.

3. Investigating Spinodal Decomposition in the Hf-Nb-Ti-V System

The sections below apply the MLIP-informed pipeline, using ORB potential energies, Jansson-based chemical potential derivatives, and a semi-implicit spectral Cahn–Hilliard solver, applied to the Hf−Nb−Ti−V system. We progress from binary to quaternary subsystems, demonstrating at each stage how thermodynamic driving forces, coherency strain, and numerical tractability jointly govern which compositions and temperatures are accessible to simulation.

Selecting a productive target composition requires meeting two criteria simultaneously. The composition must be spinodally unstable ($\lambda_{\min} < 0$ for the Gibbs energy Hessian), and no individual element should exhibit a large positive diagonal Hessian entry $H_{ii} = \partial^2 G_m^{\alpha} / \partial x_i^2 \gg 0$ at the nominal composition. Elements satisfying the latter condition are thermodynamically stable spectators: they do not drive decomposition, yet through the stability constraint on the semi-implicit spectral solver,

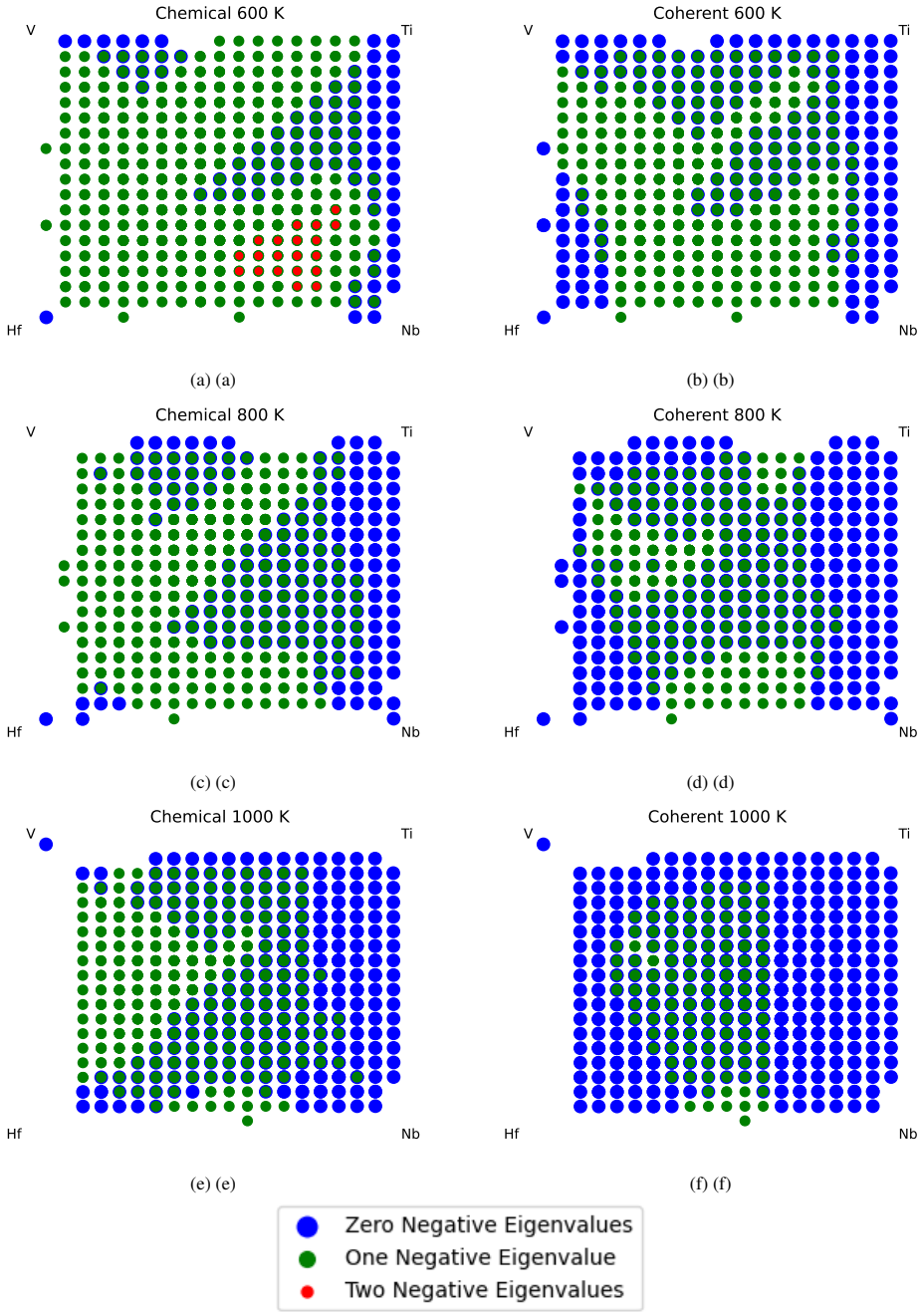
$$\Delta t \;\leq\; \frac{C_{\mathrm{s}}\,\Delta x^2}{M\,H_{\max}}, \tag{38}$$

where $C_{\mathrm{s}} = 0.3$ is a safety factor, Δx the grid spacing, M the atomic mobility, and $H_{\max}$ the largest Hessian eigenvalue, they reduce the allowable time step by one to two orders of magnitude without contributing to any spinodal driving force (Supplementary Table S1). We address both criteria by studying the binary, ternary, and quaternary subsystems in sequence, selecting compositions and temperatures at which the spinodal is both thermodynamically strong and numerically tractable.

3.1. Thermodynamic Stability Landscape

The spinodal region of the $\mathrm{BCC_{A2}}$ phase spans a wide but structured region of the Hf–Nb–Ti–V composition space whose extent and character change substantially with temperature. Figure 3 shows affine projections of the number of negative eigenvalues of the chemical and coherent Gibbs energy Hessians at 600 K, 800 K and 1000 K, computed at 5 at.% sampling density in composition space. Missing points represent convergence failures, most common at dilute compositions. At 600 K, coherent strain contributions most strongly suppress the chemical spinodal at compositions rich in Hf or V, and the region of two negative chemical eigenvalues vanishes entirely when coherency is included. At 800 K, the chemical Hessian predicts spinodal decomposition across nearly all compositions except those rich in Ti and Nb, while the coherent Hessian stabilizes solid solutions rich in both Hf and V. At 1000 K, coherent strain stabilizes all Ti- and V-rich compositions and the spinodal shrinks to a narrow Nb-rich band.

The mechanism by which coherency suppresses the Hf-rich spinodal is revealed by the first eigenvector of the chemical Hessian. Figure 4 shows the components of this vector for BCC $\mathrm{Hf_{0.6}Nb_{0.1}Ti_{0.1}V_{0.2}}$ from 600 K to 1000 K, a composition at which the chemical Hessian is negative while all eigenvalues of the coherent Hessian are positive. The unstable fluctuation mode corresponds to enrichment in Hf at the expense of V. Since Hf and V have the largest atomic radius mismatch of the four constituents, this fluctuation mode incurs the greatest coherency penalty: adding the elastic contribution to the Gibbs energy reverses the sign of the local curvature and stabilizes the alloy against this



(a) (a) (b) (b) (c) (c) (d) (d) (e) (e) (f) (f) (g)

Figure 3: Affine projection stability maps displaying the number of negative eigenvalues of the chemical and coherent Gibbs energy Hessian for BCC_{A2} in the Hf−Nb−Ti−V system at (a,b) 600 K, (c,d) 800 K, and (e,f) 1000 K. Left column: chemical Hessian; right column: coherent Hessian. Coherent strain suppresses the spinodal most strongly in Hf- and V-rich compositions across all temperatures.

mode throughout the temperature range shown. The element pair that drives the chemical instability is self-suppressed by the strain energy it would generate.

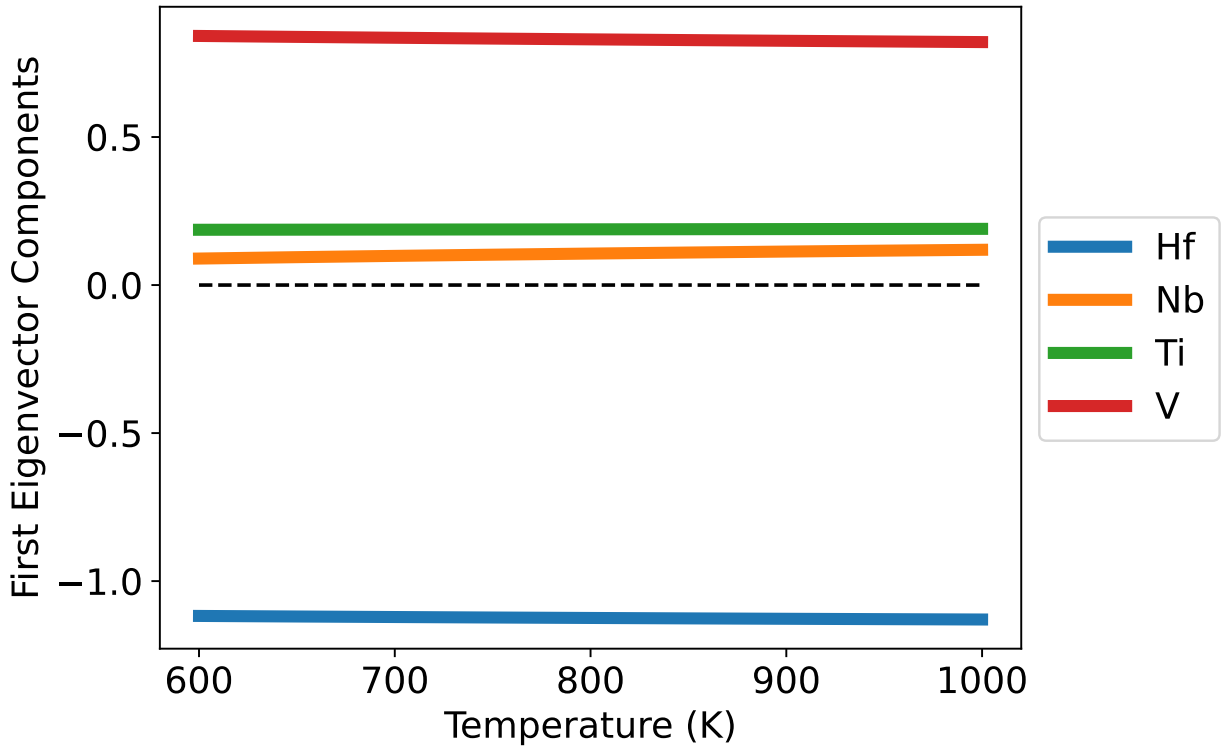
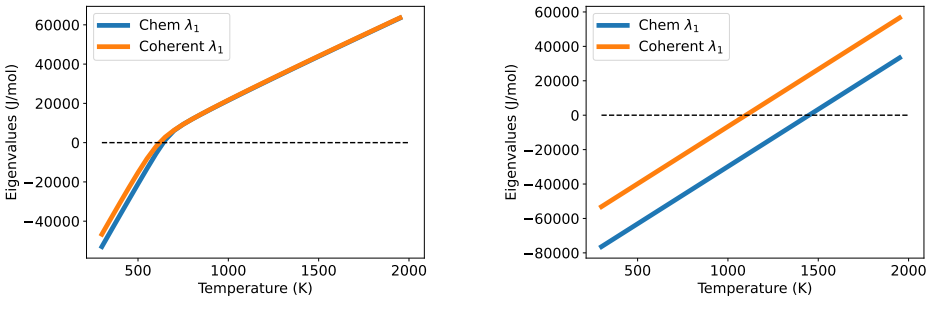


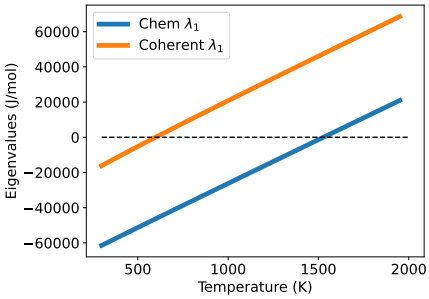
Figure 4: Components of the first gauge-invariant eigenvector of the chemical Hessian as a function of temperature for BCC $Hf_{0.6}Nb_{0.1}Ti_{0.1}V_{0.2}$. The dominant instability mode corresponds to Hf-enrichment at the expense of V; the large Hf–V atomic radius mismatch causes coherent strain to stabilize this mode at all temperatures shown, reversing the sign of the first eigenvalue from negative (chemical) to positive (coherent).

The temperature at which spinodal decomposition becomes accessible is composition-dependent and shifts substantially when coherency is included (Fig. 5). For the Ti- and Nb-rich composition $Hf_{0.10}Nb_{0.40}Ti_{0.40}V_{0.10}$, the chemical and coherent spinodal temperatures are nearly identical, reflecting the small misfit penalty in dilute-Hf alloys. For the equiatomic composition $Hf_{0.25}Nb_{0.25}Ti_{0.25}V_{0.25}$, both Hessians become negative below ~800 K, confirming that this alloy is coherently spinodal at 500 °C, the temperature adopted for all phase-field simulations below. For the Hf-rich composition $Hf_{0.6}Nb_{0.1}Ti_{0.1}V_{0.2}$, the coherent Hessian remains positive at all temperatures shown, consistent with the eigenvector analysis above.



(a) (a)

(b) (b)



(c) (c)

Figure 5: Gauge-invariant first eigenvalue of the chemical and coherent Gibbs energy Hessian as a function of temperature for BCC (a) $Hf_{0.10}Nb_{0.40}Ti_{0.40}V_{0.10}$, (b) $Hf_{0.25}Nb_{0.25}Ti_{0.25}V_{0.25}$, and (c) $Hf_{0.6}Nb_{0.1}Ti_{0.1}V_{0.2}$. Coherency shifts the spinodal temperature downward most strongly in Hf-rich alloys. The equiatomic composition (b) is coherently spinodal below ~800 K, confirming 500 °C as a tractable simulation temperature.

3.2. Binary Nb–V: First Demonstration of the MLIP-Informed Pipeline

The binary Nb–V system is the first demonstration of the complete framework: a machine-learned interatomic potential (ORB) supplies the thermodynamic database, Jansson interpolation evaluates chemical potentials and their composition derivatives directly from the CALPHAD dataset, bypassing finite-difference approximations, and the spectral Cahn-Hilliard solver propagates the resulting elasto-chemical driving forces on the grid. The single independent composition field isolates the Jansson derivative from multi-component complexity, and the well-characterized miscibility gap at 500 °C with no spectator elements provides a clean validation setting before extension to the ternary and quaternary subsystems. The equilibrium BCC_{A2} tie-line endpoints at 500 °C are an Nb-rich phase ($x_{Nb} = 0.96$) and a V-rich phase ($x_{Nb} = 0.36$), between which the stiffness tensors C_{ij} and Vegard eigenstrain coefficient $\alpha_{Nb} = \partial \ln a / \partial x_{Nb}$ are linearly interpolated from SQS configurations evaluated with the ORB machine-learned interatomic potential. The coefficient $\alpha_{Nb} = 0.0874$ is evaluated at the tie-line midpoint ($x_{Nb} = 0.66$, $a_{ref} = 3.228\,\text{Å}$), giving a total misfit strain $\varepsilon_T = -5.5\%$ between the two equilibrium phases and a molar volume $N_V = 8.99 \times 10^4\,\text{mol m}^{-3}$. Table 4 lists the full elastic constants and derived properties at both endpoints.

Table 4: Elastic constants and Vegard eigenstrain parameters for the Nb–V BCC_{A2} tie-line endpoints at 500 °C, computed from SQS configurations evaluated with the ORB machine-learned interatomic potential. $C' = (C_{11} - C_{12})/2$: tetragonal shear modulus; $A_Z = C_{44}/C'$: Zener anisotropy ratio; $\varepsilon_T = (a_{ppt} - a_{mat})/a_{mat}$: misfit strain.

Property	Nb-rich ($x_{Nb} = 0.96$)	V-rich ($x_{Nb} = 0.36$)
Composition and lattice parameter		
x_{Nb}	0.96	0.36
x_V	0.04	0.64
a (Å)	3.312	3.130
Cubic elastic constants (GPa)		
C_{11}	243.6	259.3
C_{12}	119.7	145.0
C_{44}	20.6	18.0
$C' = (C_{11} - C_{12})/2$	62.0	57.2
Derived elastic properties		
G_{VRH} (GPa)	32.5	29.2
B_{VRH} (GPa)	161.0	183.1
Poisson ratio ν	0.406	0.424
Zener ratio $A_Z = C_{44}/C'$	**0.332**	**0.315**
Vegard eigenstrain parameters (shared)		
a_{ref} (Å)	3.228	
$\alpha_{Nb} = \partial \ln a / \partial x_{Nb}$	0.0874	
$\varepsilon_T = \Delta a / a_{mat}$	−5.5%	
N_V (mol m^{-3})	8.99×10^4	

Both endpoints have $A_Z < 1$ (Table 4), identifying ⟨110⟩ as the elastically soft direction. Figure 6 shows the output of the MLIP-informed pipeline across four mean compositions. The chemical-only simulations (top row) produce isotropic interconnected domains at all four compositions, with no preferred orientation. The elasto-chemical simulations (bottom row) reveal a composition-dependent response that is absent in the chemical-only case. At $x^0_{\mathrm{Nb}} = 0.50$, coherency strain completely suppresses spinodal decomposition and the field remains uniform, consistent with the coherent Hessian analysis showing that elastic energy closes the miscibility gap at the symmetric composition. At $x^0_{\mathrm{Nb}} = 0.60$ and 0.70, decomposition proceeds and produces a labyrinthine, bicontinuous morphology with channels preferentially aligned along ⟨110⟩, reflecting the elastically soft direction ($A_Z < 1$). At $x^0_{\mathrm{Nb}} = 0.80$, the V-rich phase constitutes a small minority and forms isolated ⟨110⟩-elongated islands embedded in a continuous Nb-rich matrix.

The absence of decomposition at $x^0_{\mathrm{Nb}} = 0.50$ reflects a direct thermodynamic consequence of coherency: at this composition and temperature, the elastic penalty is sufficient to render the smallest positive eigenvalue of the coherent Gibbs energy Hessian positive, stabilizing the alloy against all composition fluctuations. That the chemical-only simulation at the same composition decomposes freely establishes that this stability is entirely elastic in origin. Coherency strain thus closes the miscibility gap at the symmetric composition in Nb–V, an effect that a purely chemical phase-field model would miss entirely.

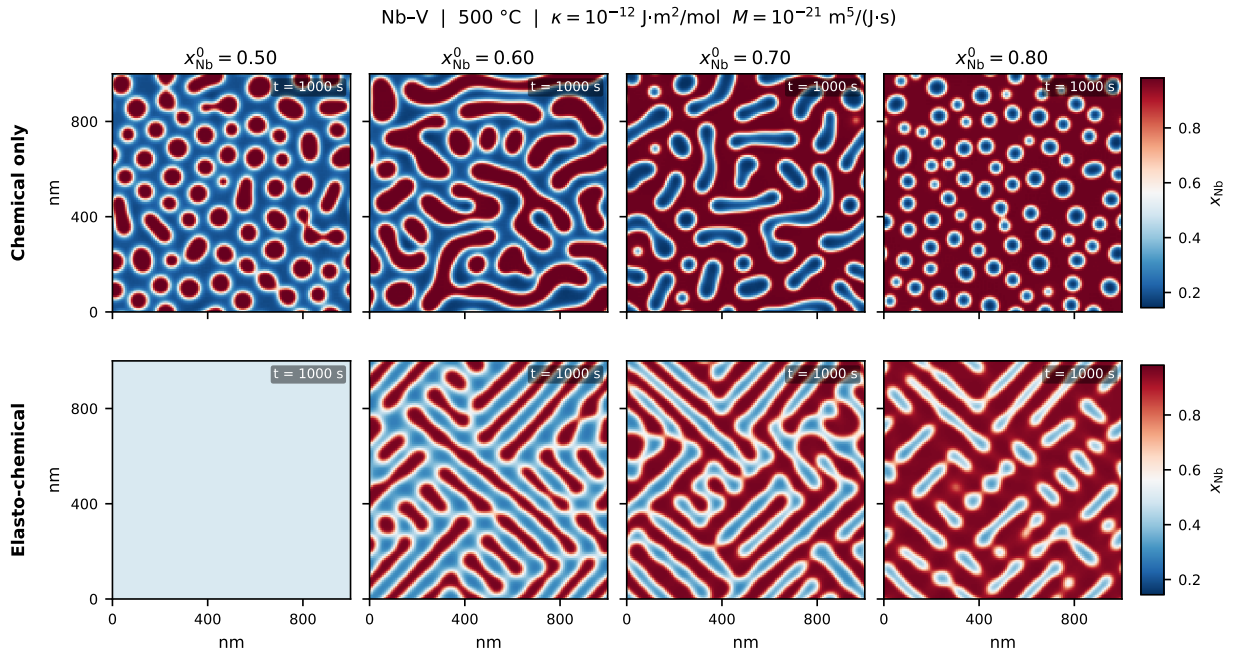


Figure 6: Simulated x_{Nb} microstructures from Cahn-Hilliard simulations of binary Nb–V $\mathrm{BCC_{A2}}$ at 500 °C (128 × 128 grid, $L = 1\,\mu\mathrm{m}$, $\kappa = 1 \times 10^{-12}\,\mathrm{J\,m^2\,mol^{-1}}$, $M = 5 \times 10^{-21}\,\mathrm{m^5\,J^{-1}\,s^{-1}}$). Top row: chemical only. Bottom row: elasto-chemical (tensorial Vegard coupling, $\alpha_{\mathrm{Nb}} = 0.087$, cubic stiffness from SQS/DFT). Columns correspond to mean compositions $x^0_{\mathrm{Nb}} = 0.50, 0.60, 0.70, 0.80$. Colormap: x_{Nb} increases from blue (low) to red (high). Both phases have $A_Z < 1$, selecting ⟨110⟩-aligned boundaries and the characteristic diamond-faceted morphology in the elasto-chemical case.

3.3. Ternary Nb–Hf–V: Temperature and Composition Dependence

Hf functions as a dual-action tuning element in the Nb–V spinodal: it deepens the chemical miscibility gap as its fraction increases, yet its large atomic radius simultaneously raises the coherency penalty that opposes decomposition. These two competing effects define a composition window, roughly $x^0_{\mathrm{Hf}} \leq 0.25$, within which spinodal microstructures are accessible under elasto-chemical coupling at 500 °C, and place a strict upper bound on Hf content above which coherency strain closes the gap entirely.

The temperature dependence compounds this picture. Table 5 identifies four regimes for $Nb_{50}Hf_5V_{45}$ from the signs of the diagonal Hessian elements H_{Nb}, H_{Hf}, and $H_{\mathrm{V}} = H_{\mathrm{Nb}} + H_{\mathrm{Hf}} + 2H_{\mathrm{Nb,Hf}}$. Above 700 °C no element is individually unstable; the spinodal survives only through weak off-diagonal Nb–Hf coupling ($\lambda_{\min} \lesssim 0$) and produces no significant microstructure. Between 600 °C and 700 °C, $H_{\mathrm{Nb}} < 0$ alone drives an asymmetric instability in which Nb segregates against a stable V+Hf background. Below 500 °C, V also destabilizes ($H_{\mathrm{V}} < 0$), recovering binary-like Nb–V demixing, but Hf then acts as a positive-curvature stiffener that severely constrains the stable time step (Eq. 38). The optimal operating point is 500 °C–700 °C, where the Nb-driven spinodal is thermodynamically active and Hf remains a spectator rather than a numerical bottleneck.

Table 5: Spinodal character and equilibrium two-phase compositions as a function of temperature for the binary $Nb_{50}V_{50}$ and ternary $Nb_{50}Hf_5V_{45}$ BCC_{A2} alloys. Signs of the diagonal Hessian elements indicate whether each element is individually unstable ($H_{ii} < 0$, driving force) or stable ($H_{ii} > 0$, spectator). H_{V} is derived via $H_{\mathrm{V}} = H_{\mathrm{Nb}} + H_{\mathrm{Hf}} + 2H_{\mathrm{Nb,Hf}}$.

System	Temperature	Instabilities	Hf role	Δt	Character	BCC phases[a]
Binary $Nb_{50}V_{50}$	T < 750 °C	$H_{\mathrm{Nb}}<0 = H_{\mathrm{V}}<0$	absent	fast	Symmetric Nb↔V demixing	BCC_1: $Nb_{99}V_1$; BCC_2: Nb_1V_{99}
Ternary $Nb_{50}Hf_5V_{45}$	T > 700 °C	$H_{\mathrm{Nb}}>0$, $H_{\mathrm{Hf}}>0$, $H_{\mathrm{V}}>0$	stable	fast	No instability; marginal off-diagonal Nb–Hf coupling only	Single-phase BCC
	600 °C to 700 °C	$H_{\mathrm{Nb}}<0$, $H_{\mathrm{Hf}}>0$, $H_{\mathrm{V}}>0$	spectator	moderate	Asymmetric Nb-driven spinodal; optimal window	BCC_1: $Nb_{90}Hf_9V_1$; BCC_2: $Nb_2Hf_{0.1}V_{98}$
	T < 500 °C	$H_{\mathrm{Nb}}<0$, $H_{\mathrm{Hf}}>0$, $H_{\mathrm{V}}<0$	stiffener	penalized	Binary-like Nb–V demixing; $H_{\mathrm{Hf}}>0$ constrains Δt	BCC_1: $Nb_{91}Hf_9V_{0.3}$; BCC_2: $Nb_{0.6}Hf_{0.04}V_{99}$

[a] Tie-line endpoints from `pycalphad` (`BCC_A2`, ORB-MLIP database); subscripts at.%. 500 °C for T<750 °C rows, 700 °C for 600 °C to 700 °C row.

Figure 7 consolidates three further perspectives on the ternary system. Figure 7(a) shows the chemical-only Hf-content sweep across six compositions (x^0_{Hf} = 0.05–0.45), each after 6000 s of isothermal aging at 500 °C. At low Hf fractions (x^0_{Hf} = 0.05, 0.10) decomposition produces isolated Nb-rich precipitates consistent with a minority-phase microstructure. Increasing Hf to 0.20 and 0.25 shifts the tie-line endpoints and produces well-developed bicontinuous Nb–V domains, indicating that Hf progressively deepens the miscibility gap. At x^0_{Hf} = 0.35 the domains coarsen to a coarser, lower-contrast pattern, and in $Nb_{50}Hf_{45}V_5$ no decomposition occurs, placing this composition outside the chemical spinodal at 500 °C.

Figure 7(b) shows the corresponding elasto-chemical sweep using the composition-dependent coherency eigenstrain model; snapshots are taken after 30 000 s for $Nb_{50}Hf_5V_{45}$, $Nb_{50}Hf_{10}V_{40}$, $Nb_{50}Hf_{20}V_{30}$, and $Nb_{50}Hf_{25}V_{25}$; 15 000 s for $Nb_{50}Hf_{35}V_{15}$; and 150 000 s for $Nb_{50}Hf_{45}V_5$. The elastic coupling further narrows the spinodal region: $Nb_{50}Hf_5V_{45}$, $Nb_{50}Hf_{10}V_{40}$, $Nb_{50}Hf_{20}V_{30}$, and $Nb_{50}Hf_{25}V_{25}$ still decompose with phase contrast and domain connectivity similar to the chemical-only case, while $Nb_{50}Hf_{35}V_{15}$ and $Nb_{50}Hf_{45}V_5$ remain uniform throughout the simulation. The suppression at $x^0_{Hf} = 0.35$ is purely elastic in origin: the chemical-only simulation at the same composition decomposes, whereas coherency strain raises the effective free-energy curvature above zero and stabilizes the solid solution. This trend, where elastic stabilization becomes dominant as Hf content increases, is consistent with the larger Vegard misfit parameter of Hf relative to V and the attendant increase in elastic self-energy at Hf-rich compositions. Figure 7(c) maps the simulated compositions onto the ternary BCC_{A2} miscibility gap, projected across the Hf–Nb, Nb–V, and Hf–V binary subsystems.

3.4. Quaternary HfNbTiV: Phase-Field Prediction and Experimental Validation

Equiatomic $Nb_{25}Hf_{25}Ti_{25}V_{25}$ lies within the BCC_{A2} spinodal at 500 °C ($\lambda_{min} < 0$) and produces well-evolved microstructure on accessible simulation timescales. The chemical-only simulation at 60 000 s yields ellipsoidal V-rich precipitates ($x^{max}_V = 0.751$) in an Nb/Hf/Ti-rich matrix (Figure 8(a)). Diffuse Hf halos around precipitate cores arise from the tenfold lower Hf mobility rather than an independent Hf instability, consistent with the kinetic analysis in Section 3.3.

The elasto-chemical simulation at 700 000 s reproduces the same V-rich/Nb-Ti-rich partitioning but with reduced compositional contrast ($x^{max}_V = 0.691$), reflecting the elastic penalty that opposes sharp composition gradients (Figure 8(b)). CALPHAD equilibrium predicts two coexisting BCC_{A2} phases ($f_{ppt} \approx 0.33$, Table 6). Both simulations approach the CALPHAD endpoints for Nb ($x^{mat}_{Nb} \approx 0.30$) and V ($x^{ppt}_V \approx 0.71$), while Hf and Ti remain far from their equilibrium values, consistent with slow Hf kinetics and the limited simulation timescales accessible.

Table 6: CALPHAD tie-line endpoints for equiatomic $Nb_{25}Hf_{25}Ti_{25}V_{25}$ BCC_{A2} at 500 °C (ORB-MLIP database), with chemical and elasto-chemical phase-field composition ranges shown for comparison. PFM values are non-equilibrium, particularly for Hf and V.

		x_{Nb}	x_{Hf}	x_{Ti}	x_V
CALPHAD tie-line	Matrix ($f \approx 0.67$)	0.300	0.371	0.305	0.024
	Precipitate ($f \approx 0.33$)	0.148	0.002	0.136	0.714
Chemical PFM (60 000 s)	Matrix (max)	0.331	0.365	0.310	0.751
	Precipitate (min)	0.096	0.048	0.103	0.015
Elasto-chem. PFM (700 000 s)	Matrix (max)	0.396	0.426	0.353	0.691
	Precipitate (min)	0.127	0.056	0.125	0.053

The MLIP-computed elastic constants predict a spinodal modulation direction in the quaternary that differs qualitatively from the binary and ternary baselines: the SQS/ORB-MLIP interpolant evaluated at the CALPHAD matrix

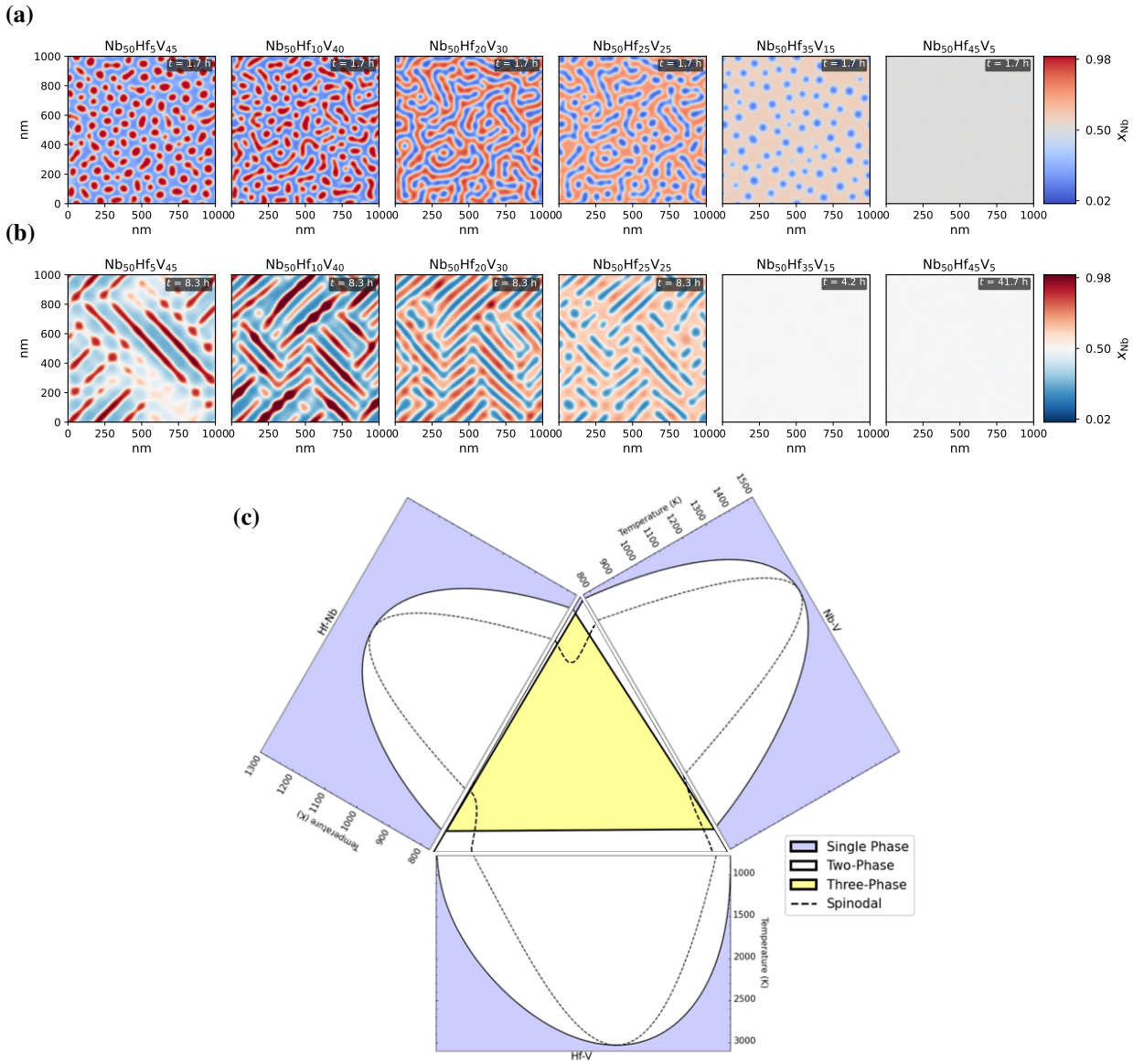


Figure 7: Hf-content sweep of x_{Nb} microstructures in ternary Nb–Hf–V $\mathrm{BCC_{A2}}$ at 500 °C (128 × 128 grid, $L = 1\,\mu\mathrm{m}$, $\kappa = 1 \times 10^{-12}\,\mathrm{J\,m^2\,mol^{-1}}$, $M_{\mathrm{Nb}} = 5 \times 10^{-21}\,\mathrm{m^5\,J^{-1}\,s^{-1}}$). **(a)** Chemical-only simulations, all compositions after 6000 s of isothermal aging. **(b)** Elasto-chemical simulations using the composition-dependent coherency eigenstrain strain model; snapshots after 30 000 s ($\mathrm{Nb_{50}Hf_5V_{45}}$, $\mathrm{Nb_{50}Hf_{10}V_{40}}$, $\mathrm{Nb_{50}Hf_{20}V_{30}}$, $\mathrm{Nb_{50}Hf_{25}V_{25}}$), 15 000 s ($\mathrm{Nb_{50}Hf_{35}V_{15}}$), and 150 000 s ($\mathrm{Nb_{50}Hf_{45}V_5}$). Compositions with $x^0_{\mathrm{Hf}} \geq 0.35$ show no spinodal decomposition, demonstrating that coherency strain suppresses the miscibility gap at high Hf content. **(c)** Ternary $\mathrm{BCC_{A2}}$ miscibility gap across all three binary subsystems (Hf–Nb, Nb–V, Hf–V). Colormap: x_{Nb} increases from blue (low) to red (high).

composition yields $A_Z^{\mathrm{mat}} = 1.52 > 1$, selecting ⟨100⟩ as the elastically soft direction and therefore the preferred microstructure alignment, opposite to the binary NbV ($A_Z = 0.33 < 1$) and ternary NbHfV ($A_Z = 0.37 < 1$) cases where ⟨110⟩ is favored. Table 7 compares the full set of model predictions against the experimental characterization of equiatomic HfNbTiV by An et al. [3], who identified two coexisting $\mathrm{BCC_{A2}}$ phases in an as-cast alloy by XRD,

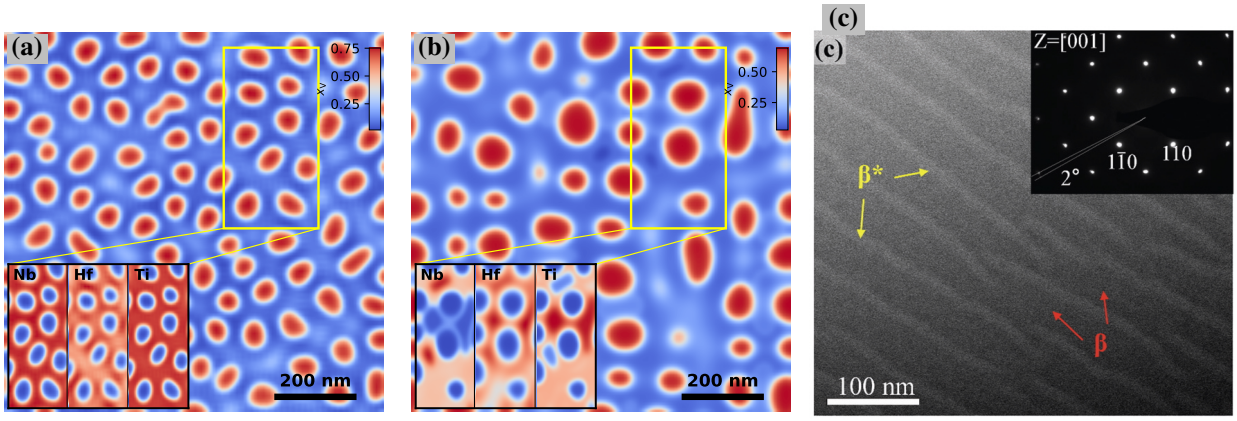


Figure 8: Phase-field simulations of equiatomic $Nb_{25}Hf_{25}Ti_{25}V_{25}$ at 500 °C and HAADF-STEM comparison. (a) Chemical-only simulation at 60 000 s: x_V field (128 × 128, $L = 1\,\mu m$) with insets showing x_{Nb}, x_{Hf}, x_{Ti} for the highlighted region. (b) Elasto-chemical simulation at 700 000 s: same fields with elastic coupling. (c) HAADF-STEM image of as-cast alloy showing alternating β/β^* BCC_{A2} variants; reproduced from An et al. [3].

HAADF-STEM, and EDS, revealing three areas of agreement and disagreement.

Table 7: Predicted and experimentally measured properties for the equiatomic BCC_{A2} HfNbTiV alloy. Experimental values are from An et al. [3] (as-cast, characterized by XRD, HAADF-STEM, and EDS). Binary predictions use the $Nb_{50}V_{50}$ CALPHAD tie-line endpoints (matrix: $Nb_{96}V_4$; precipitate: $Nb_{36}V_{64}$) at 500 °C. Ternary predictions use the $Nb_{50}Hf_{25}V_{25}$ CALPHAD tie-line endpoints (matrix: $Nb_{70}Hf_{10}V_{20}$; precipitate: $Nb_{15}Hf_{35}V_{50}$) at 500 °C. Quaternary predictions use the equiatomic $Nb_{25}Hf_{25}Ti_{25}V_{25}$ CALPHAD tie-line at 500 °C (Table 6). $A_Z = 2C_{44}/(C_{11} - C_{12})$: $A_Z > 1$ selects ⟨100⟩ modulation, $A_Z < 1$ selects ⟨110⟩. $\varepsilon_T = (a_{ppt} - a_{mat})/a_{mat}$.

Property	Binary $Nb_{50}V_{50}$ (this work)	Ternary $Nb_{50}Hf_{25}V_{25}$ (this work)	Quaternary $Nb_{25}Hf_{25}Ti_{25}V_{25}$ (this work)	Experiment [3]
Precipitate enriched in	V	V	V (Hf in matrix)	Hf + V
Matrix enriched in	Nb	Nb	Nb, Hf, Ti	Nb + Ti
Precipitate phase fraction	—	—	≈ 0.33	≈ 1/3
a_{mat} (Å)	3.312	3.286	3.382	3.313
a_{ppt} (Å)	3.130	3.265	3.089	3.259
ε_T	−5.5%	−0.64%	−8.7%	−1.6% (XRD); 0.54% (TEM)
A_Z (matrix)	0.33	0.37	**1.52**	—
A_Z (precipitate)	0.31	1.09	0.32	—
Spinodal modulation direction	⟨110⟩ (predicted)	⟨110⟩ (predicted)	⟨100⟩ (predicted)	⟨110⟩ (observed)
Modulation wavelength	—	—	≈50–100 nm	50–100 nm

Hf partitioning is reversed between the CALPHAD model and experiment. An et al. demonstrate via HAADF-STEM Z-contrast that Hf ($Z = 72$, the heaviest constituent) enriches the precipitate phase (β^*), co-segregating with V. The ORB-MLIP CALPHAD model assigns Hf to the matrix alongside Nb and Ti (Table 6). The most probable explanation is non-equilibrium partitioning during rapid solidification: the experimental alloy was drop-cast, so decomposition occurred on cooling from the melt rather than at 500 °C. Database inaccuracies in the ORB-MLIP model for heavy-element BCC alloys cannot be excluded, but the thermal-history explanation is supported by the low

equilibrium diffusivity of Hf, which makes kinetic trapping plausible.

The coherency misfit is overestimated, directly tracing to the Hf partitioning error. An et al. report a lattice misfit of −1.6% from XRD peak separation and +0.54% from TEM atomic-column measurements, both substantially smaller than the −8.7% predicted by the quaternary CALPHAD tie-line. If Hf (large atomic radius, $a_{\mathrm{Hf}}^{\mathrm{BCC}} \approx 3.62\,\text{Å}$) co-partitions with V (small radius, $a_{\mathrm{V}}^{\mathrm{BCC}} \approx 3.03\,\text{Å}$) into the precipitate, their opposing contributions to the precipitate lattice parameter partially cancel, yielding a net misfit consistent with experiment. This internal consistency supports the thermal-history origin of the discrepancy; the binary NbV misfit of −5.5% is similarly overestimated, though less severely.

The quaternary elastic anisotropy does not correctly predict the observed ⟨110⟩ spinodal modulation direction. The SAED pattern in An et al. (Fig. 1(c) inset, [001] zone axis) shows diffraction spots along ⟨110⟩, and the β^* lamellae visible in the HAADF-STEM image are elongated along ⟨110⟩, identifying this as the experimental modulation direction. The quaternary CALPHAD model gives $A_Z^{\mathrm{mat}} = 1.52 > 1$, which predicts ⟨100⟩ modulation, inconsistent with the observed ⟨110⟩. The binary NbV and ternary NbHfV models ($A_Z < 1$) correctly predict ⟨110⟩, though for the wrong physical reason (dilute chemistry rather than the full quaternary matrix composition). Interpolating the SQS/ORB-MLIP elastic constants along a compositional path from the ternary matrix ($Nb_{70}Hf_{10}V_{20}$, $A_Z = 0.37$) to the quaternary matrix ($Nb_{30}Hf_{37}Ti_{31}V_2$, $A_Z = 1.52$) reveals that A_Z crosses unity at $x_{\mathrm{Ti}} \approx 0.13$–$0.16$, identifying Ti addition as the element responsible for flipping the predicted modulation direction from ⟨110⟩ to ⟨100⟩ (Figure 9). Ti has a high C_{44}/C' ratio in BCC that preferentially stiffens the relevant shear mode, driving A_Z above unity as its concentration increases. This discrepancy may reflect the as-cast, non-equilibrium character of the experimental microstructure, where kinetics and thermal history rather than the equilibrium elastic anisotropy govern the early-stage modulation direction.

Despite these discrepancies, the model correctly predicts a phase fraction of $\approx 1/3$ and a modulation wavelength of ≈50–100 nm, both in close agreement with the experimental observations of An et al.

The quantitative discrepancies observed in the quaternary system point primarily to limitations of the current MLIP description. The SQS/ORB-MLIP elastic constants are 0 K static estimates that have not been validated against DFT or experimental measurements for the specific quaternary compositions predicted by the CALPHAD tie-line; errors in these values propagate directly into both the coherency misfit and the A_Z prediction. Moreover, the 0 K static approximation does not capture finite-temperature lattice softening or the effect of short-range chemical order on the shear moduli, both of which are significant in multicomponent BCC alloys near a miscibility gap. Non-equilibrium solidification in the as-cast specimen may contribute to the Hf partitioning discrepancy, but the systematic nature of the elastic errors suggests that dedicated DFT validation of the quaternary elastic constants and an improved finite-temperature MLIP description are the most impactful paths toward quantitative agreement with experiment.

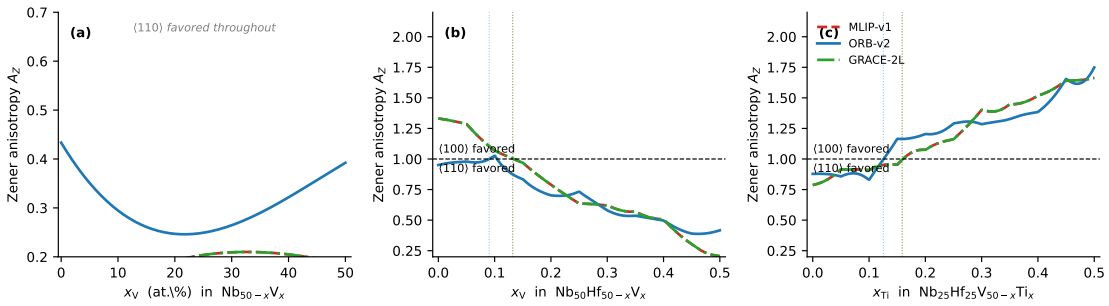


Figure 9: Zener anisotropy $A_Z = C_{44}/C'$ from SQS/MLIP elastic constants for three MLIP versions along three composition paths. (a) Binary $Nb_{50-x}V_x$: $A_Z < 1$ throughout (⟨110⟩ favored). (b) Ternary $Nb_{50}Hf_{50-x}V_x$: A_Z crosses unity near $x_V \approx 0.11$. (c) Quaternary $Nb_{25}Hf_{25}V_{50-x}Ti_x$: Ti drives the crossover at $x_{Ti} \approx 0.13$–0.16, switching modulation from ⟨110⟩ to ⟨100⟩.

4. Conclusions and Future Work

In this work, we demonstrated an open-source workflow for high-throughput microstructure stability analysis. Firstly, if a suitable CALPHAD model is not available to the modeler, it can be created through PhaseForge. The chemical Hessian is then determined analytically and efficiently via Jansson derivatives. Coherent strain contributions can be added through an analytically differentiable bond model and MLIP estimates for elastic constants. Lastly, following a clever change of basis, gauge-invariant eigenvalues and eigenvectors of the chemical and coherent spinodal can be investigated at specific compositions as functions of temperature or visualized at a given temperature over a high-dimensional composition space through affine projections. This workflow provides many valuable extensions to the foundational stability analysis framework produced by Koneru, Kadirvel, and Wang [11], to include 1) open-source access, 2) whitebox CALPHAD model knowledge, 3) improved efficiency and robustness of Gibbs energy derivatives, 4) modeling coherent strain contributions, and 5) visualization in large composition spaces via one two-dimensional plot.

However, there are still plenty of future directions to further extend this workflow. A model for homogeneous elastic Gibbs energy contributions could be incorporated in addition to contributions from coherent strain due to composition fluctuations. MLIPs could also be used to create bond models for other crystal structures, and coherent strain elastic contributions could be modeled for materials that do not behave isotropically. Lastly, implementation of second-order Jansson derivatives would provide analytic calculation of third-order derivatives of the Gibbs energy with respect to composition, enabling the analytic calculation of critical points.

Code Availability

The source code and workflows associated with this work are openly available at `https://github.com/vahid2364/PhaseFoundry`.

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